KING SALMAN INTERNATIONAL UNIVERSITY

Faculty of Medicine

Histology

Lecture -2: Cytology-II

Dr. Sarwat L.A. Abdel-Latif

2026/2027

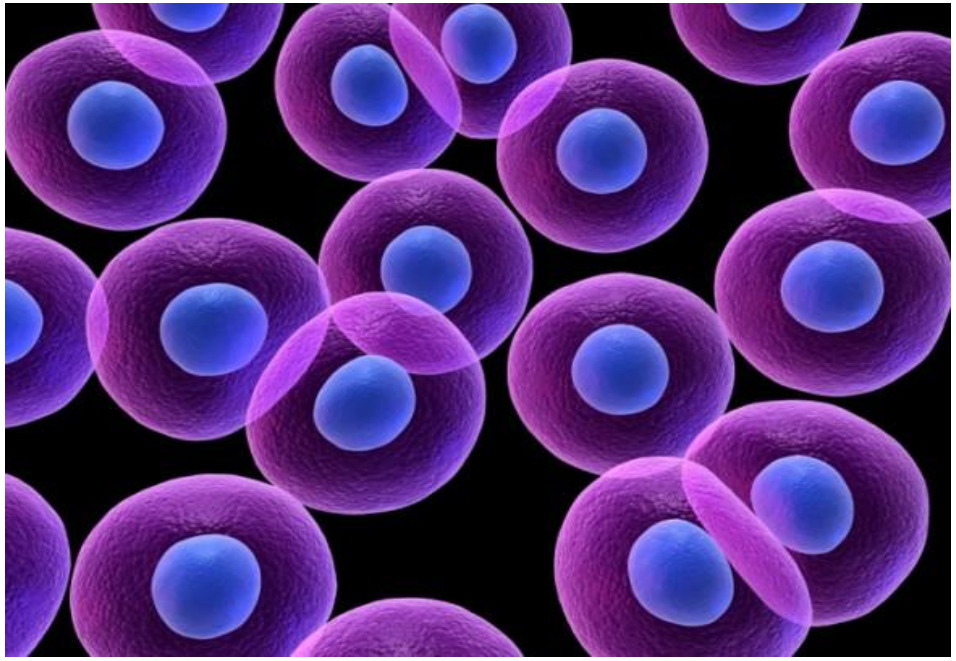
Objectives

- **List** the components of the cell, and types of membranous and non-membranous organelles.
- **Describe** the structure (LM & EM) correlated to function of endoplasmic reticulum
- **Compare** between rough and smooth endoplasmic reticulum.
- **Correlate** between the structure (L/M & E/M), and the function of Golgi apparatus
- **Define** the Lysosomes, and enumerate types
- **Describe** the structures correlated to the functions of lysosomes
- **Define and classify** cell inclusions
- **Describe** the structure and correlated functions of cell inclusions.

The Cell

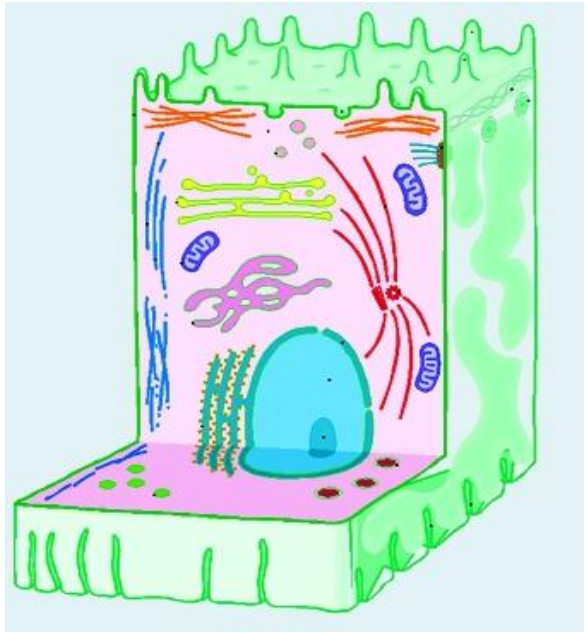
Cytoplasm

Nucleus



Cytoplasm

- Cytoplasmic Matrix
- Cytoplasmic organelles
- Cytoplasmic inclusions



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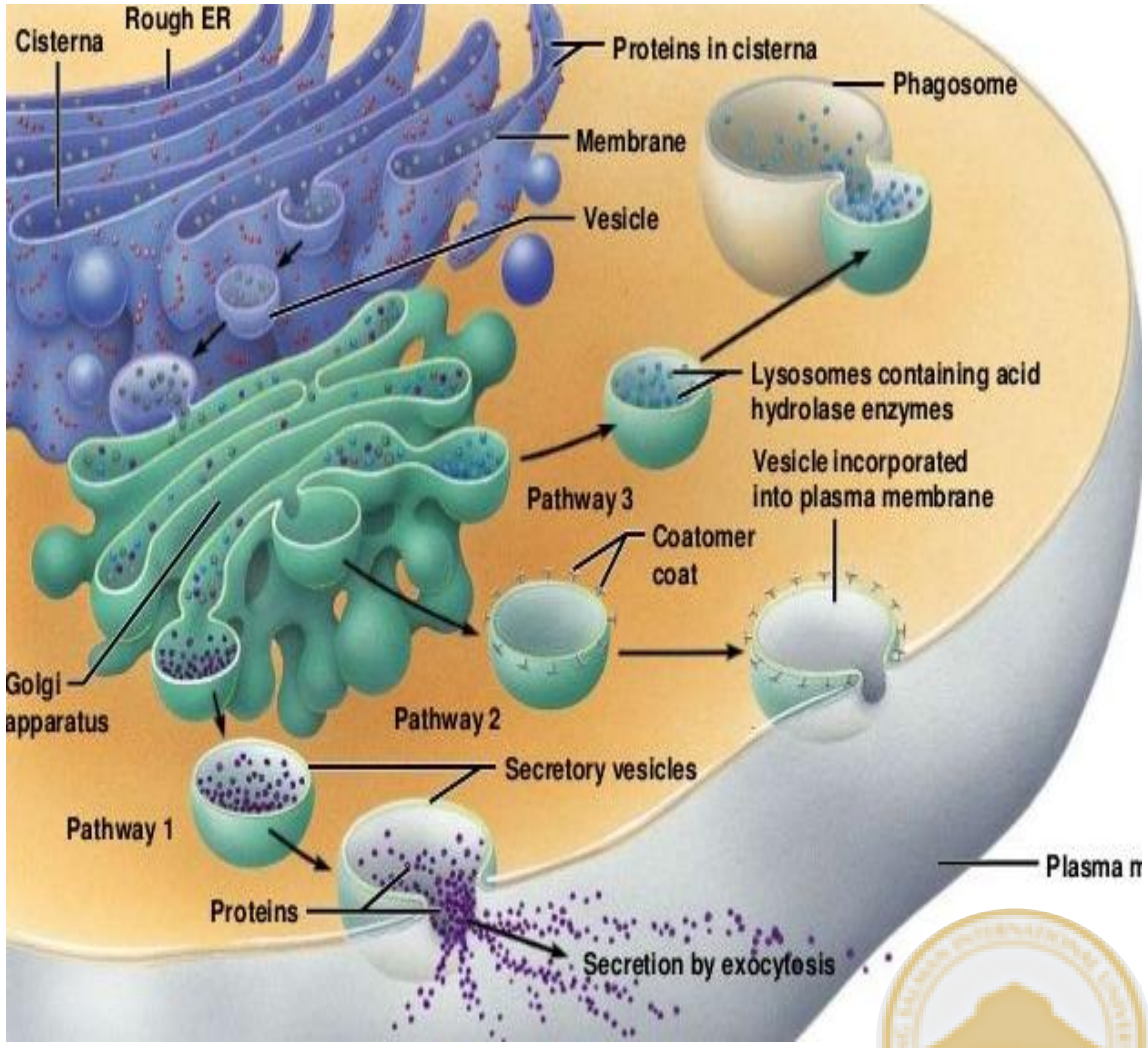
Types of Cytoplasmic Organelles

Membranous	Non-membranous
✓ Cell membrane	✓ Ribosomes
✓ Mitochondria	✓ Cytoskeleton: a-Microtubules (Centrioles & Cilia) b-Filaments (Thin, Intermediate & Thick)
✓ Endoplasmic reticulum (rough & smooth)	
✓ Golgi apparatus	
✓ Lysosomes	

Lysosomes

■**Def:**

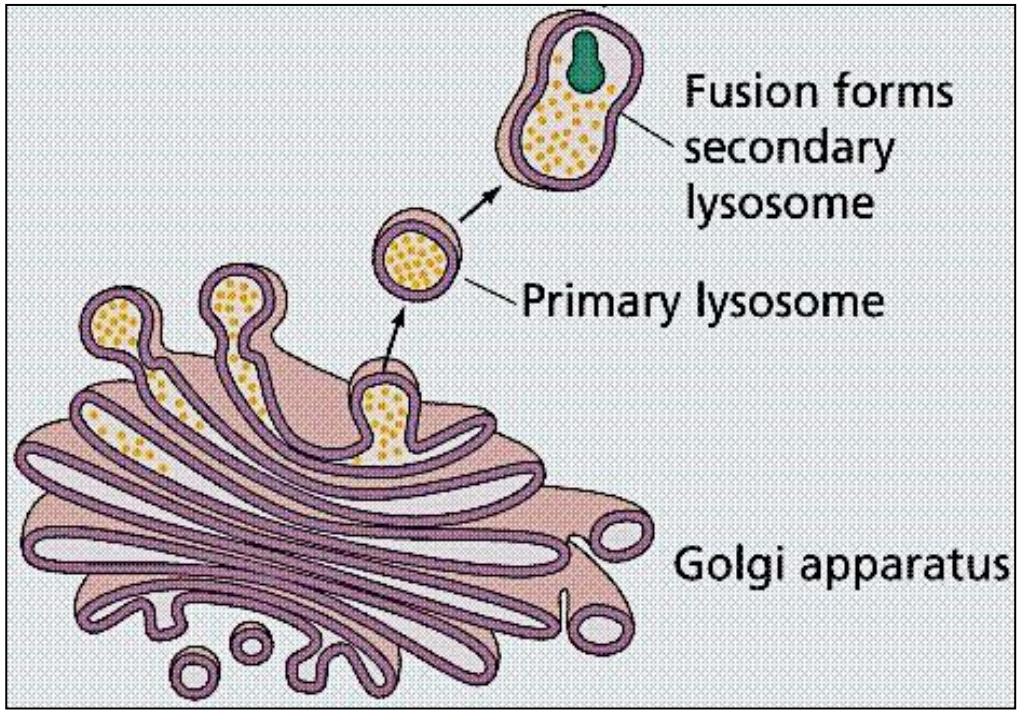
- **Digestive system of the cell**
- **Formed by rER and Golgi apparatus**
- **Contain many hydrolytic enzymes [e.g. phospholipases, proteases, nucleases**
- **Abundant in phagocytic cells [WBCs]**



Lysosomes

■E.M:

- ✓ Electron-dense appearance
- ✓ 2 Types are distinguished



➢1ry lysosomes:

✓Newly formed

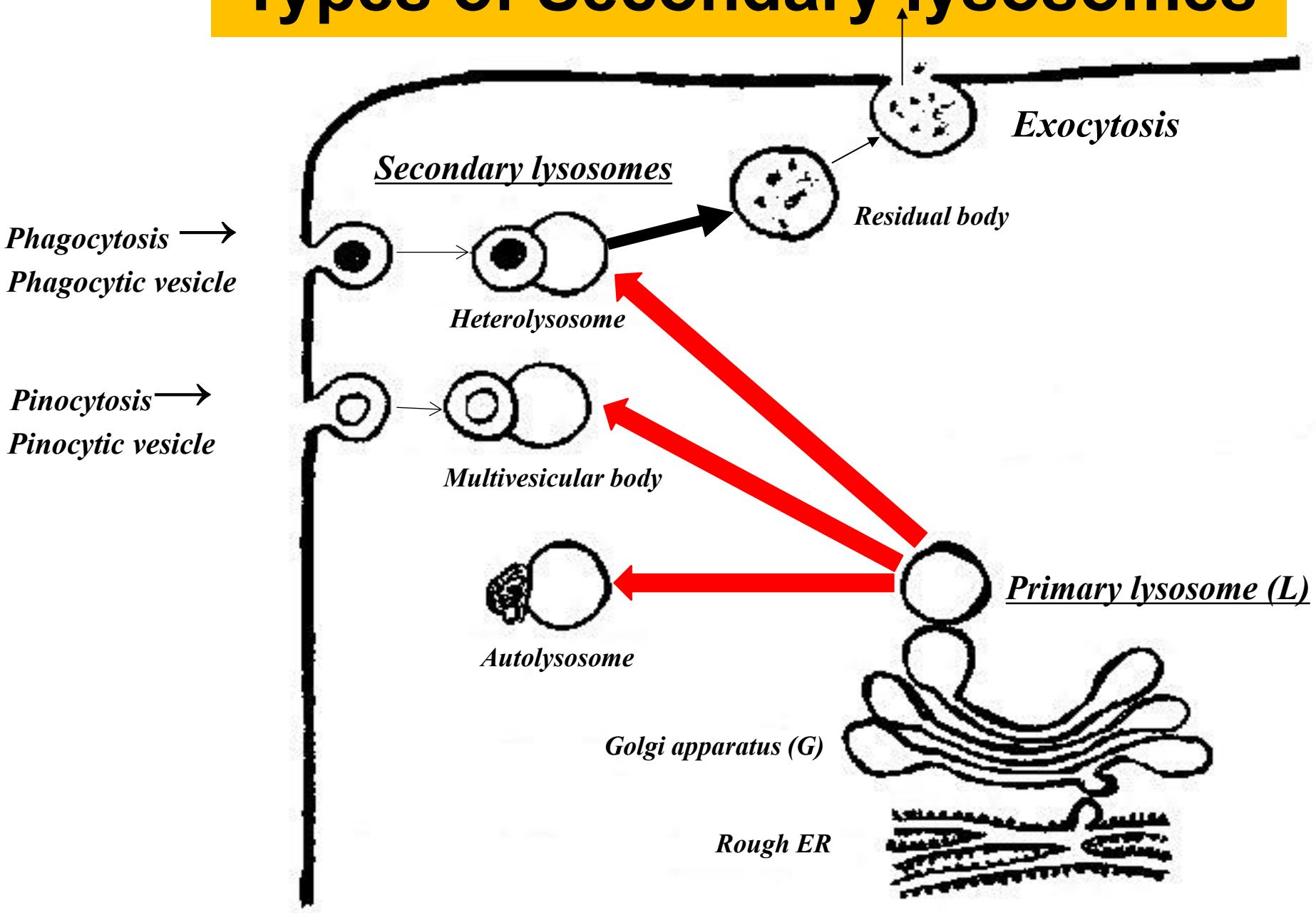
✓Small homogenous vesicles

➢2ry lysosomes:

✓Fusion of 1ry Lysosomes + intracellular molecules

✓Heterogeneous vesicles

Types of Secondary lysosomes



Types of Secondary lysosomes

- **Heterolysosomes:**

  Fusion of 1ry lysosome + engulfed solid food or bacteria

- **Multivesicular bodies:**

  Fusion of 1ry lysosome + engulfed fluid droplets

- **Autolysosomes:**

  Fusion of 1ry lysosome + destroyed old organelles [dead mitochondria, rER…]

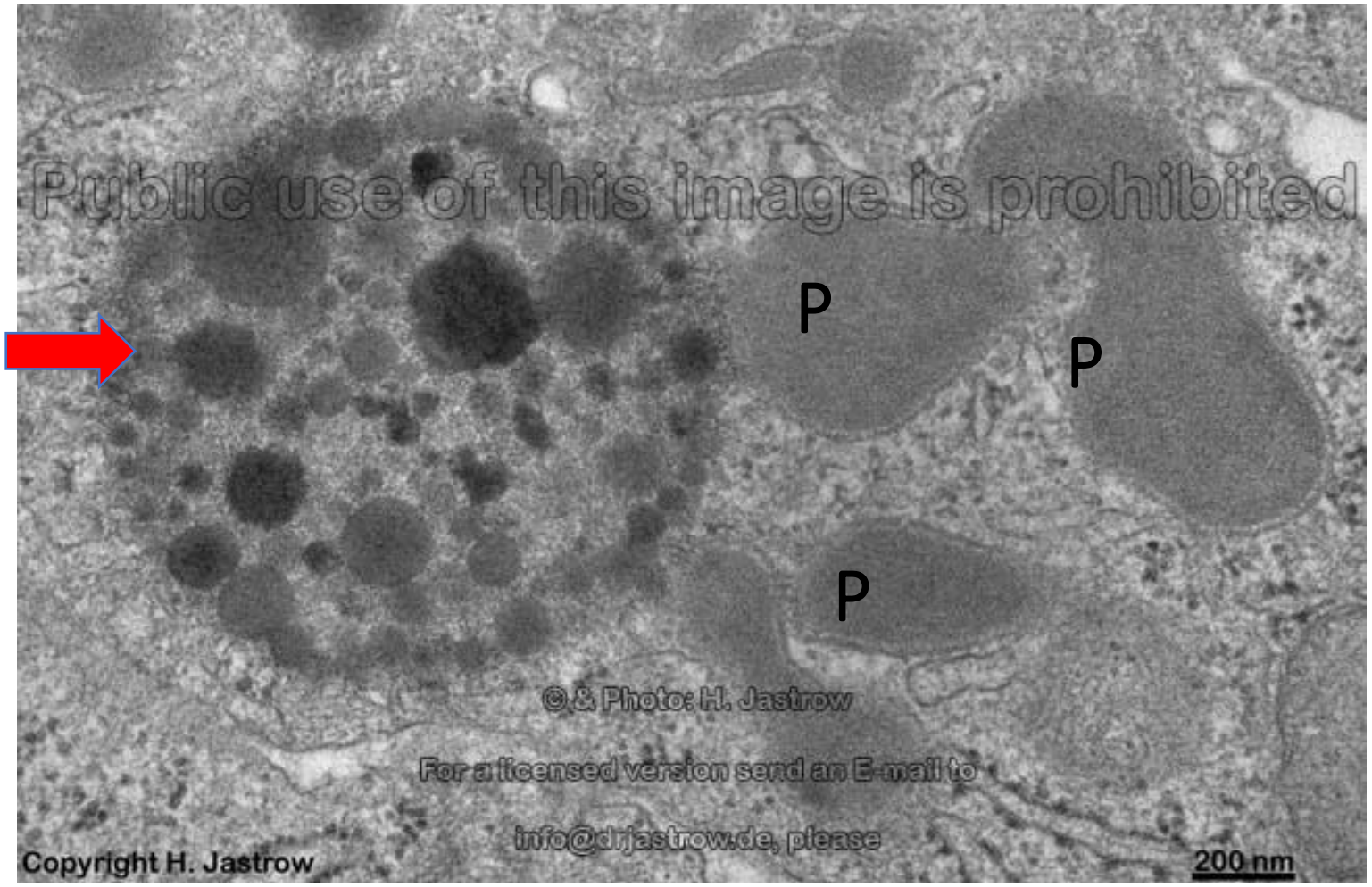
This process is called autophagy

- **Residual bodies:**

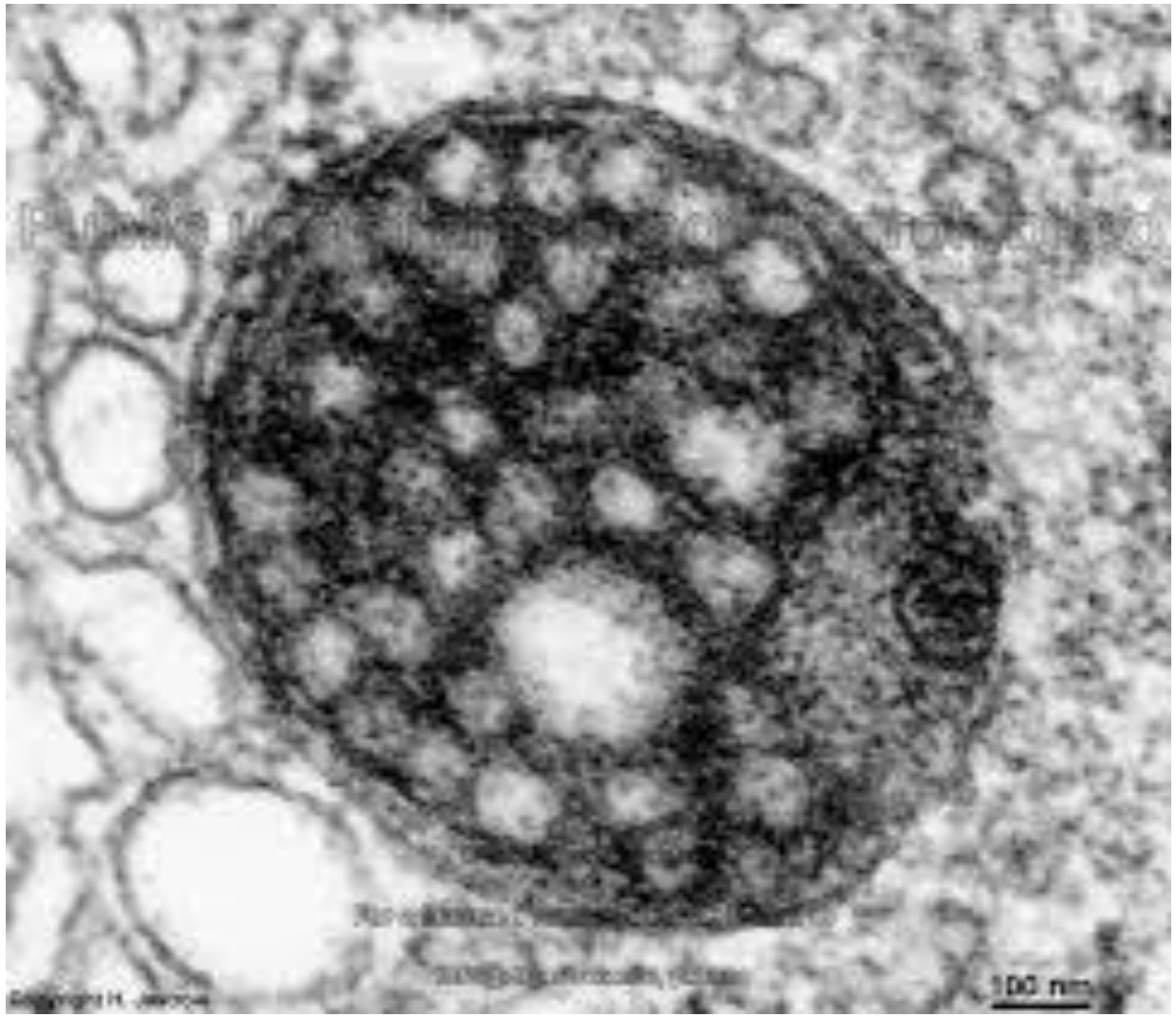
  2ry lysosomes containing indigestible remnants



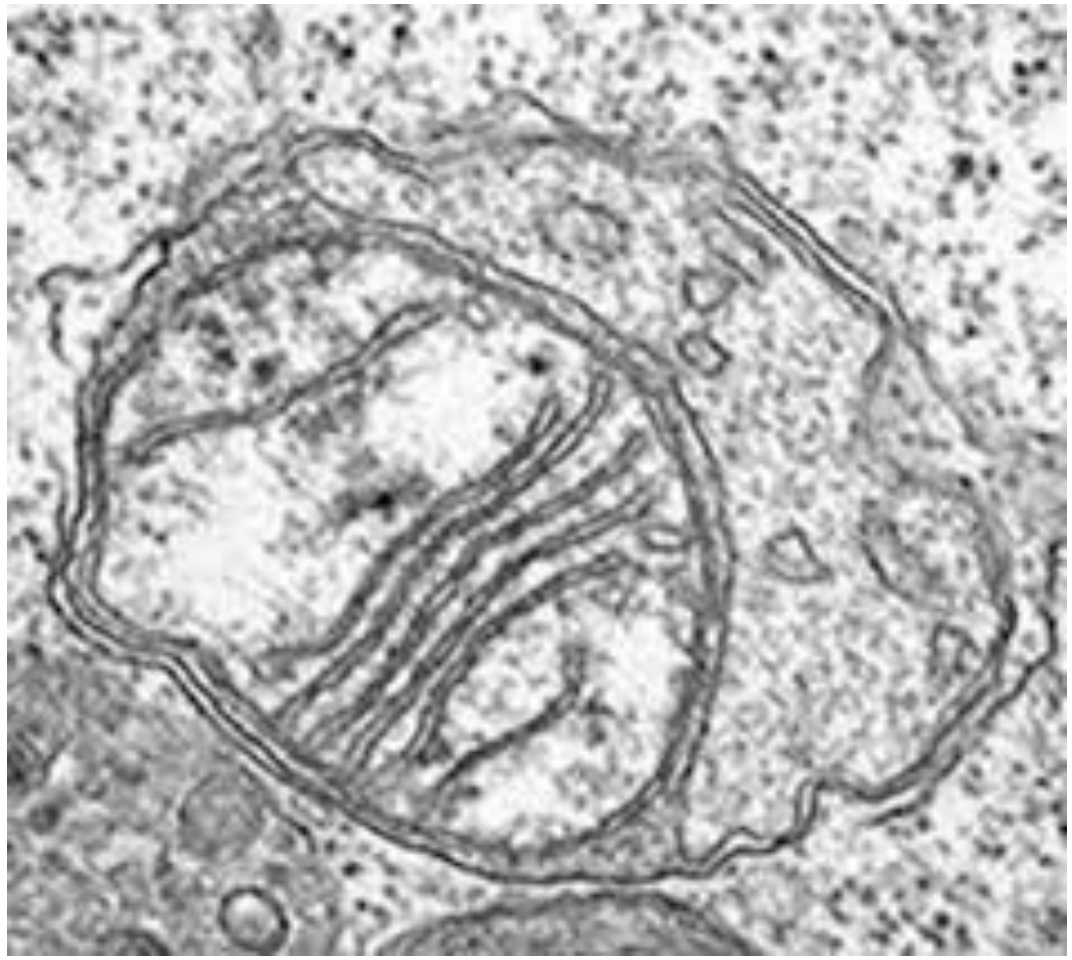
EM of Heterolysosome



EM of Multivesicular Body



EM of Autolysosome



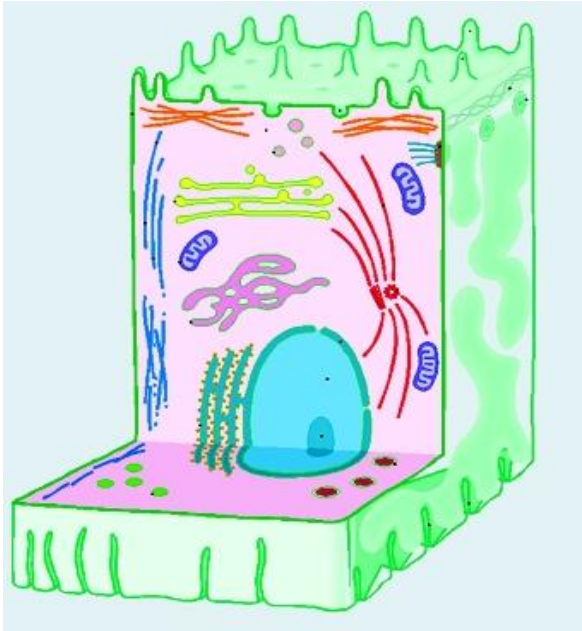
Functions of Lysosomes

- **Digestion of:**
  - ✓**Nutrients** in cytoplasm
  - ✓**Foreign particles**: bacteria & viruses → **immune function**
  - ✓**Fluid droplets**
  - ✓**Old organelles**
  - ✓**The whole cell** (autolysis after programmed cell-death)

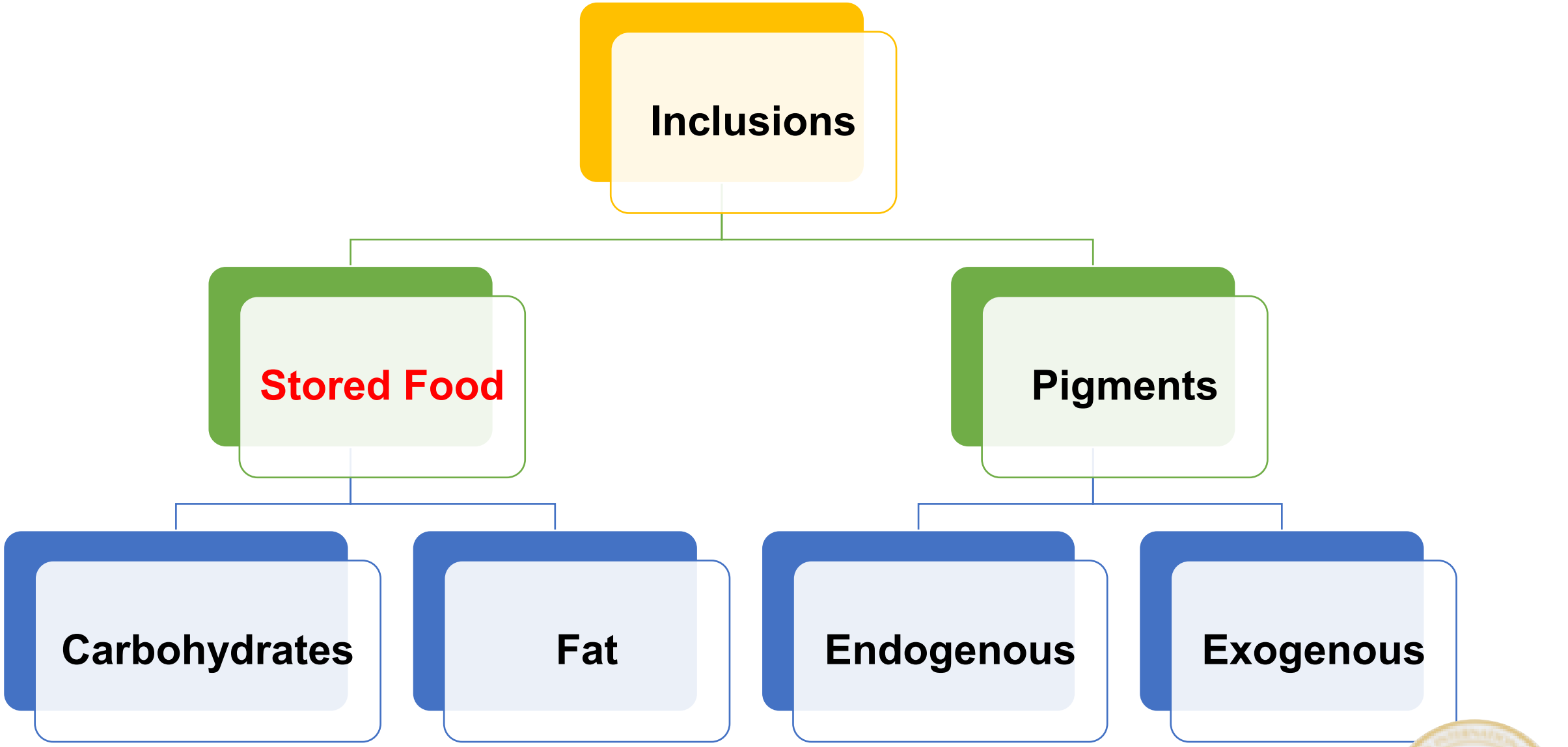
Cytoplasm

Cytoplasmic Matrix

Cytoplasmic organelles



Cytoplasmic inclusions


Inclusions
Stored Food
Pigments
Carbohydrates
Fat
Endogenous
Exogenous

Carbohydrates [Glycogen]

Site: in liver & Muscles

L.M:

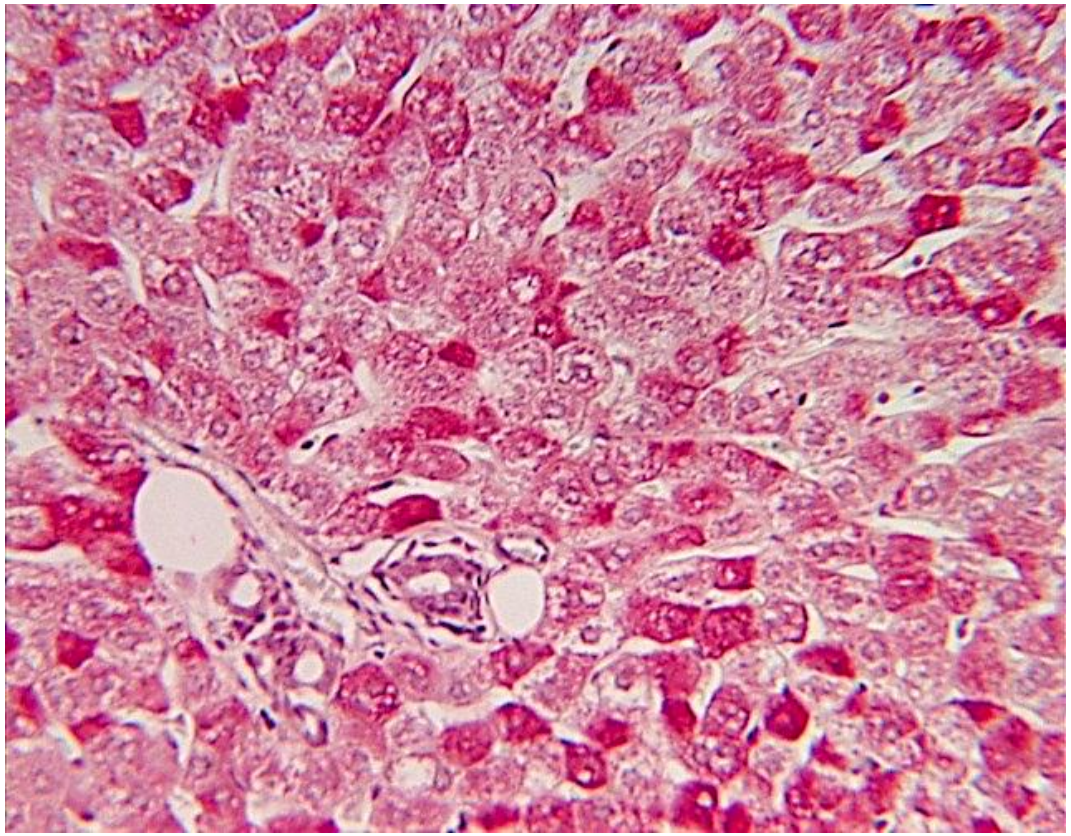
-**Not** visible with H&E [water soluble]

-Stained by

- **Best's Carmine** (red granules)

or

- **PAS** (purple).



Carbohydrates [Glycogen]

E.M:

-Electron-dense granules

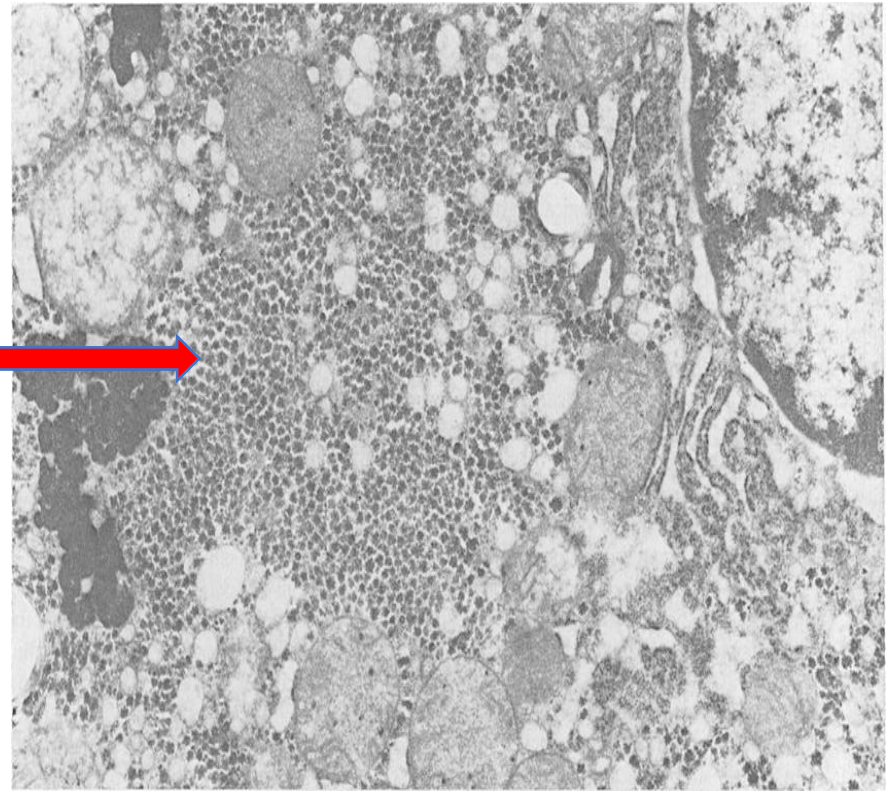
-Either single [α Granules]

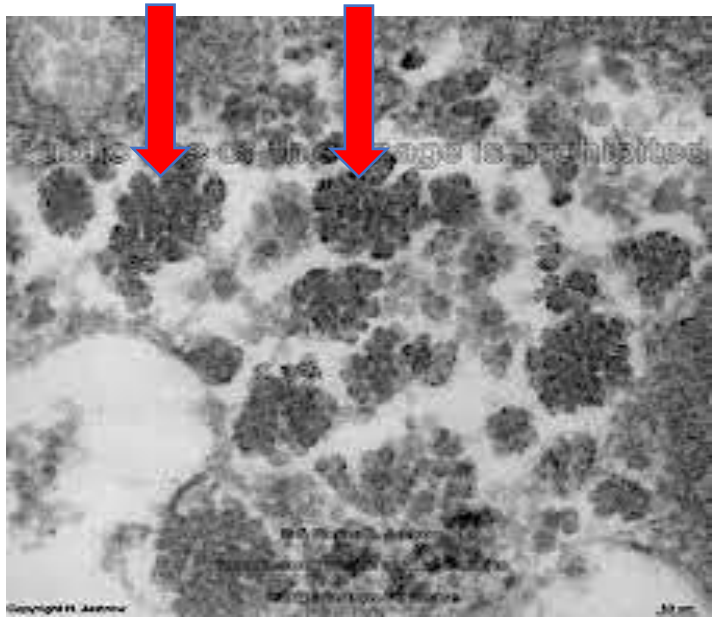
or in rosette-shaped groups [β Granules]

-Mostly concentrated in

cytoplasm in areas rich in

sER



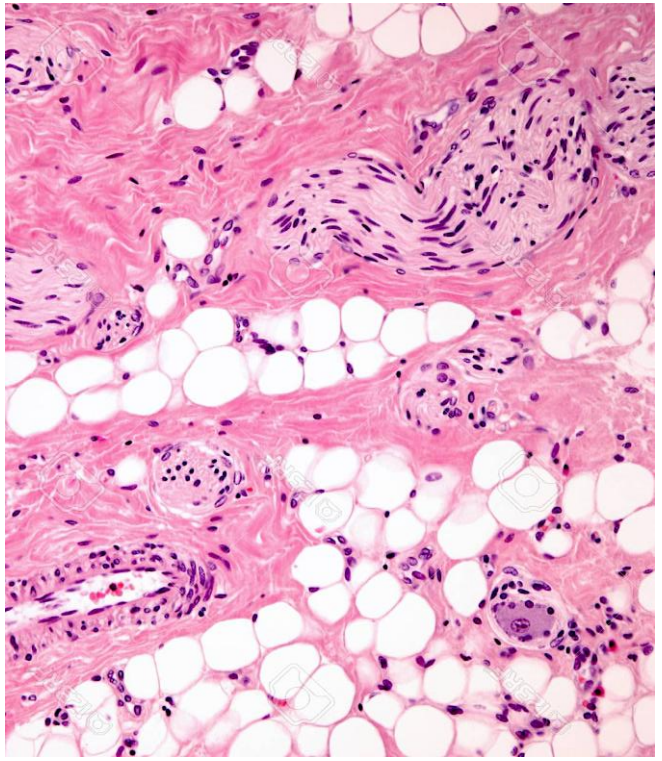


Fats [Lipid]

Sites: Fat cells & Liver

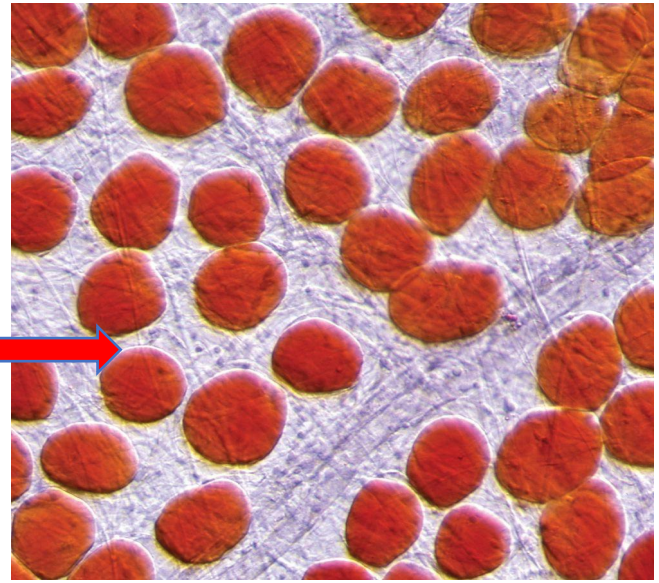
L.M:

- ✓ In H&E appear as empty vacuoles [dissolve during preparation]

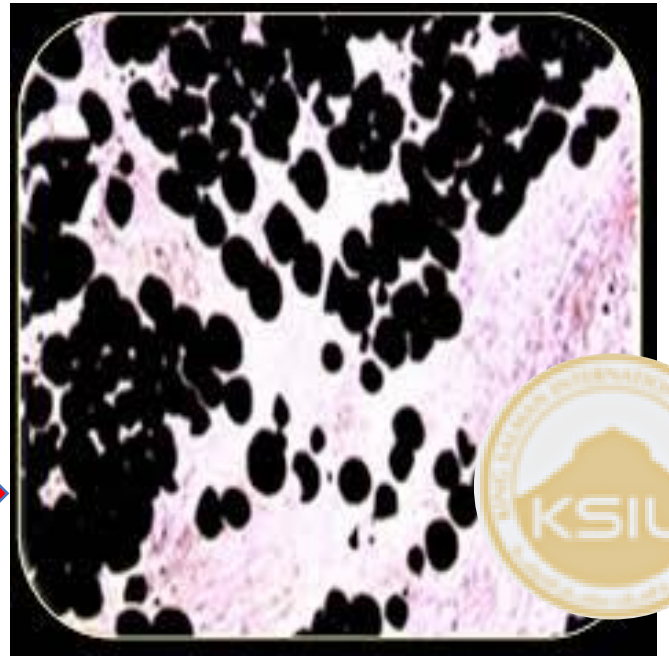


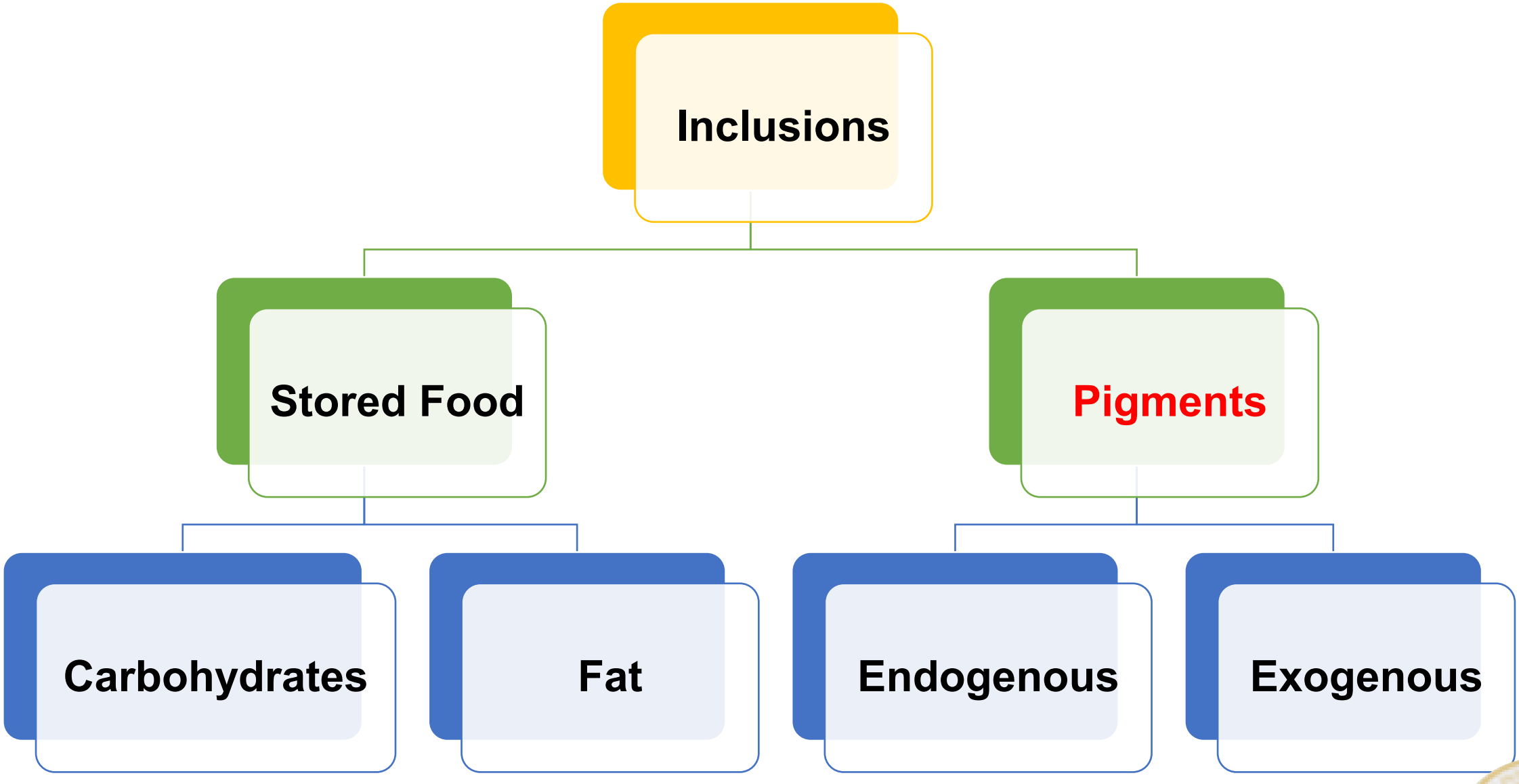
- ✓ Stained with **special stains**:

Sudan III → Orange vacuoles



Sudan black → Black vacuoles




Inclusions
Stored Food
Pigments
Carbohydrates
Fat
Endogenous
Exogenous

Endogenous Pigments

a- Hemoglobin: in RBCs.

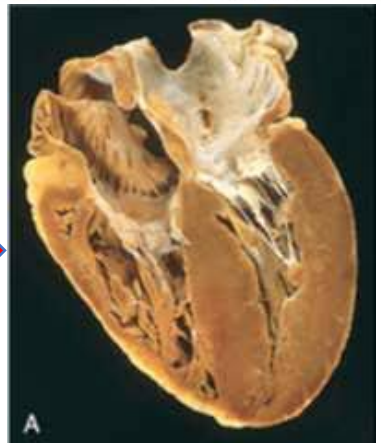


b- Melanin: in skin, hair & eye.



c- Lipofuscin granules: in long lived cells

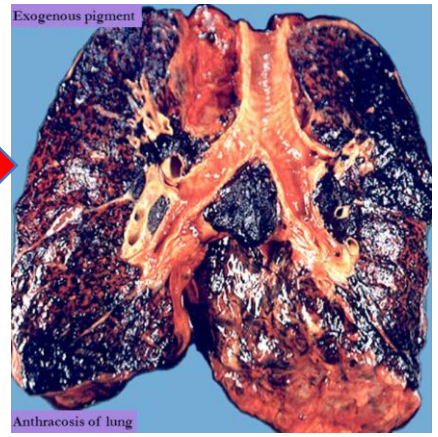
[nerve cells & heart muscle]



Exogenous Pigments

1-Carbon & Dust particles in inspired air:

Phagocytosed by dust cells → Black lungs



2-Carotene pigment in carrots →

Yellow fat in cells



3-Tattoo marks:

Dyes as India ink is injected under skin

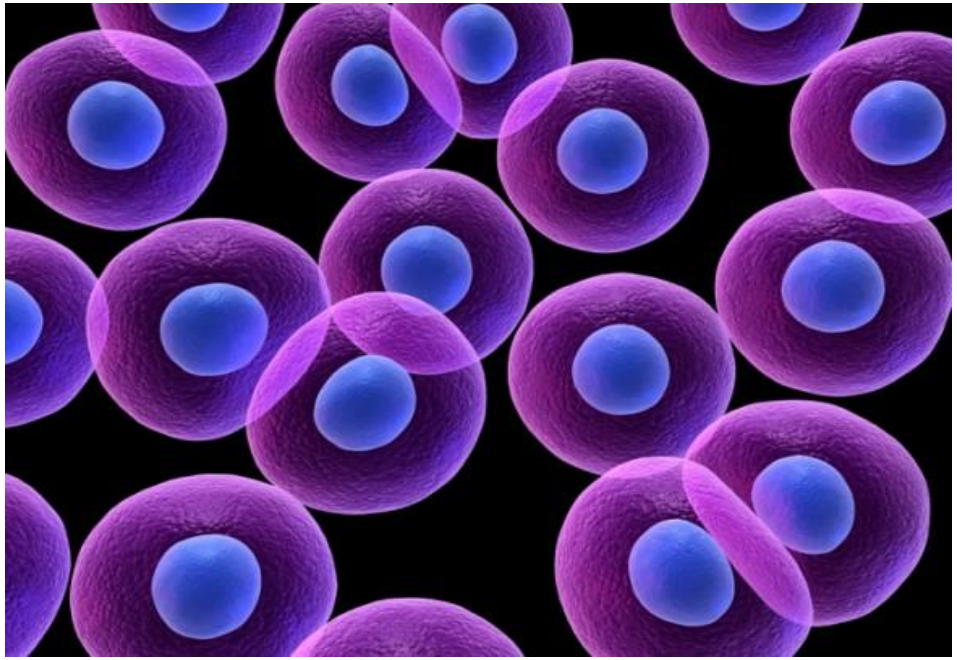
& taken by phagocytic cells



The Cell

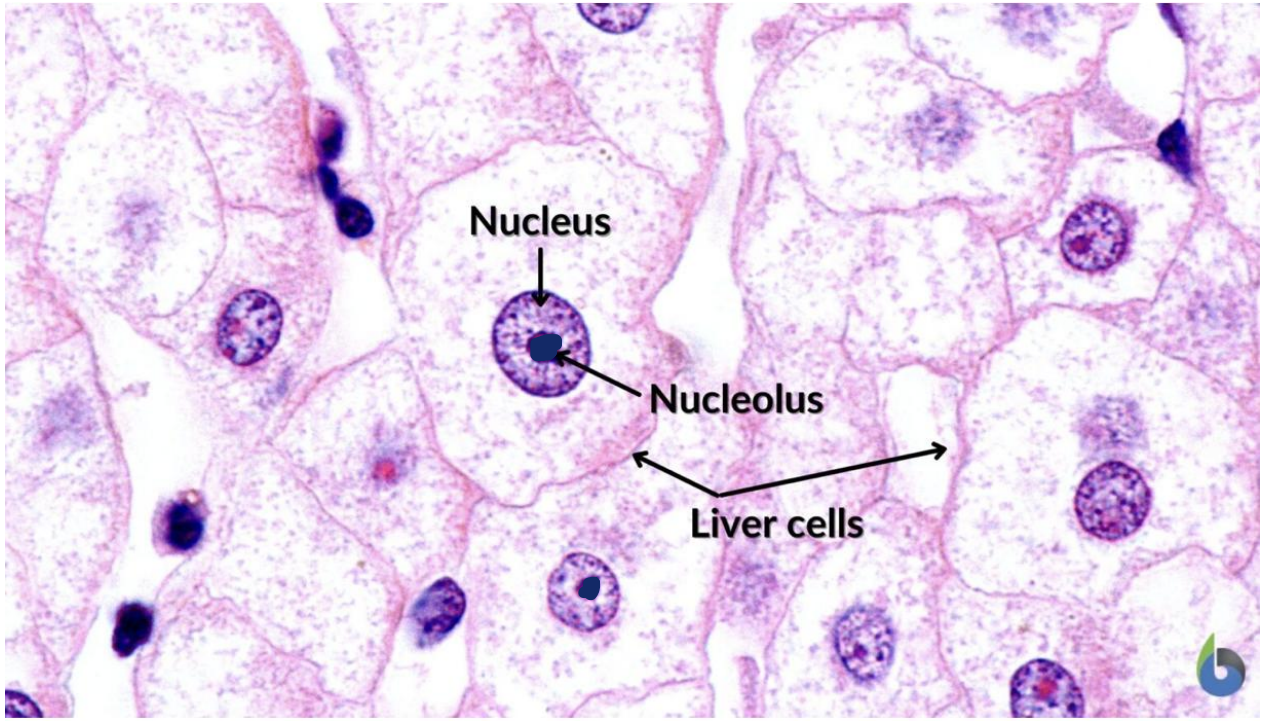
Cytoplasm

Nucleus



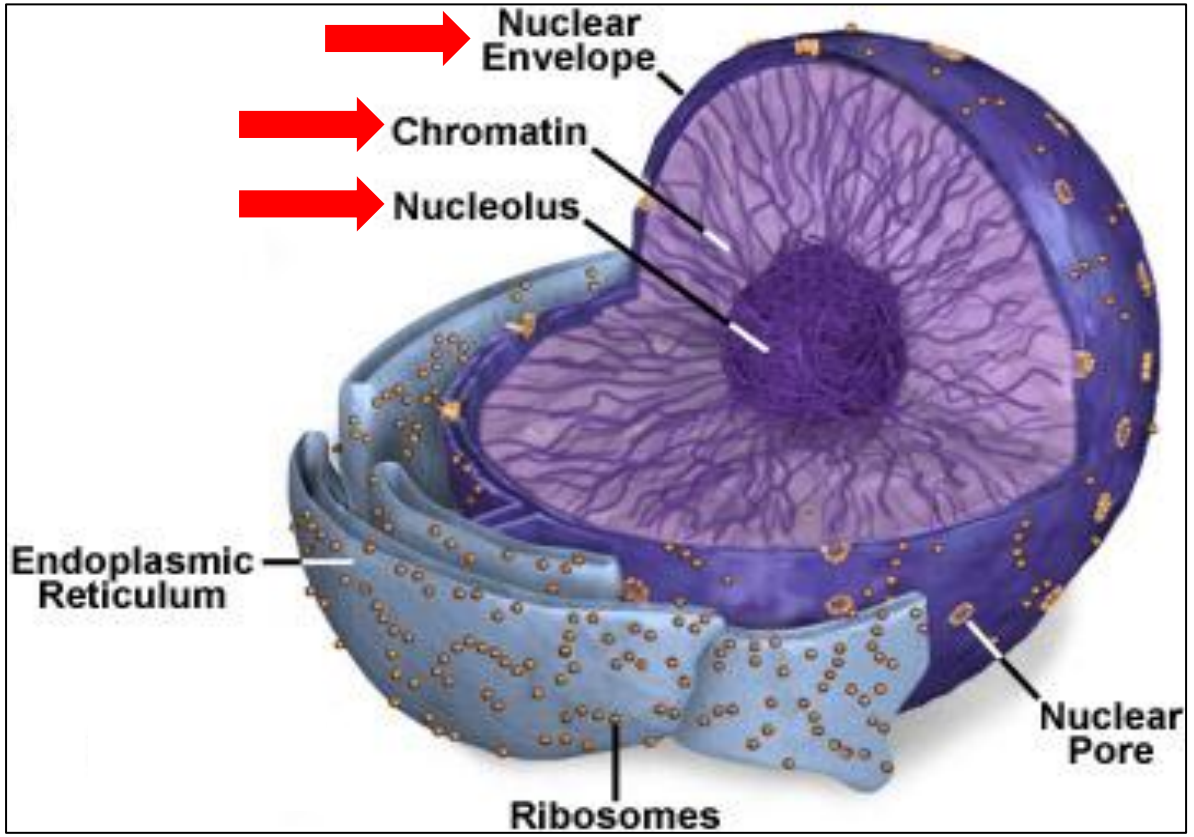
The Nucleus

- The command center of the cell
- L.M: Basophilic structure due to nucleic acids (DNA & RNA)



Structure of the Nucleus [E.M]

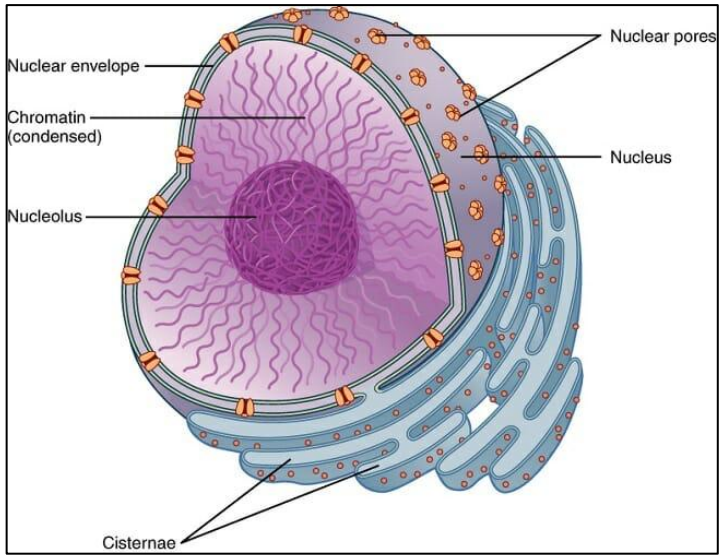
- Nuclear Envelope
- Nucleoplasm
- Chromatin
- Nucleolus

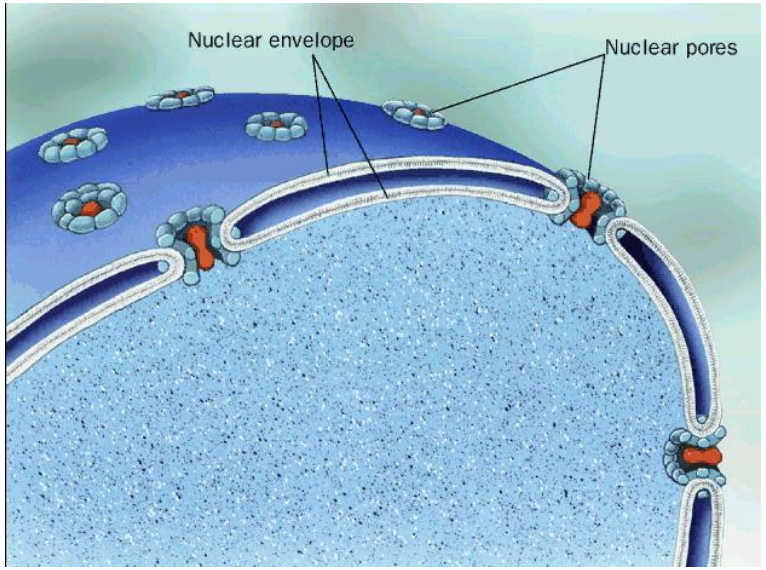


1-Nuclear Envelope

■ **E.M:**

- **Double membrane separated by perinuclear space [30-50 nm]**
- **Outer membrane & perinuclear space are continuous with rER**
- **Outer & Inner membranes fuse at intervals forming nuclear pores [3000-4000 pores in one growing cell]**
- **Nuclear pores are closed by diaphragms**
- **Nuclear pores regulate passage of substances between nucleus & cytoplasm in both directions**





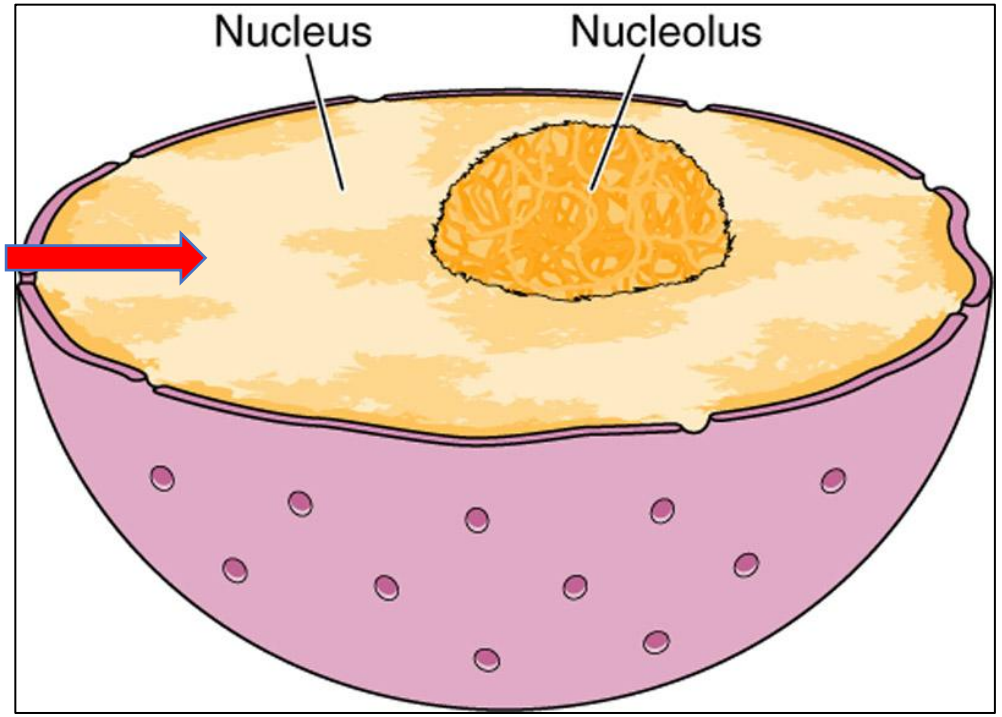
2-Nucleoplasm [Nuclear sap]

■The clear homogenous ground substance of the nucleus

■Acts as transport medium

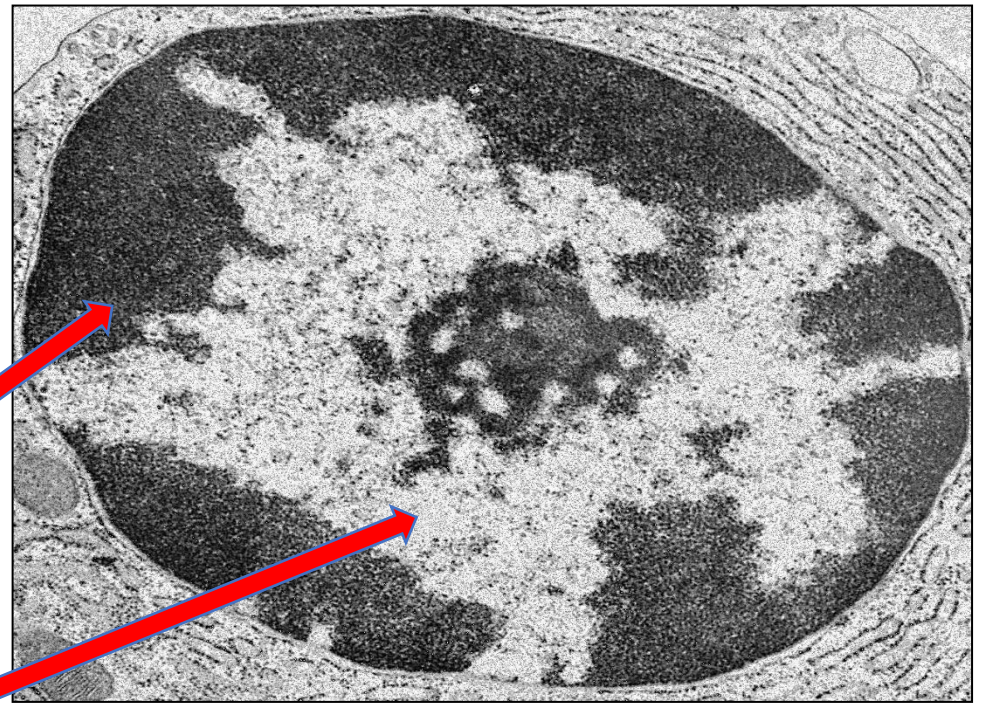
→ Carry ribosomes and m-RNA from nucleus to cytoplasm

Nucleoplasm

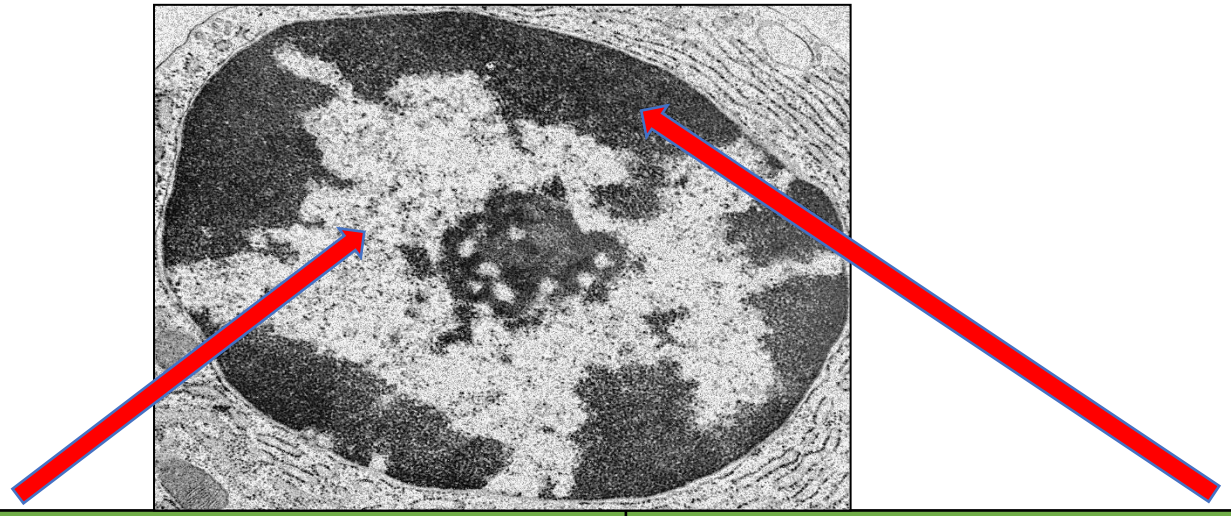


3-Chromatin

- The genetic material
- Formed of nucleoprotein
- Forms chromosomes during cell division



Types of chromatin by E.M

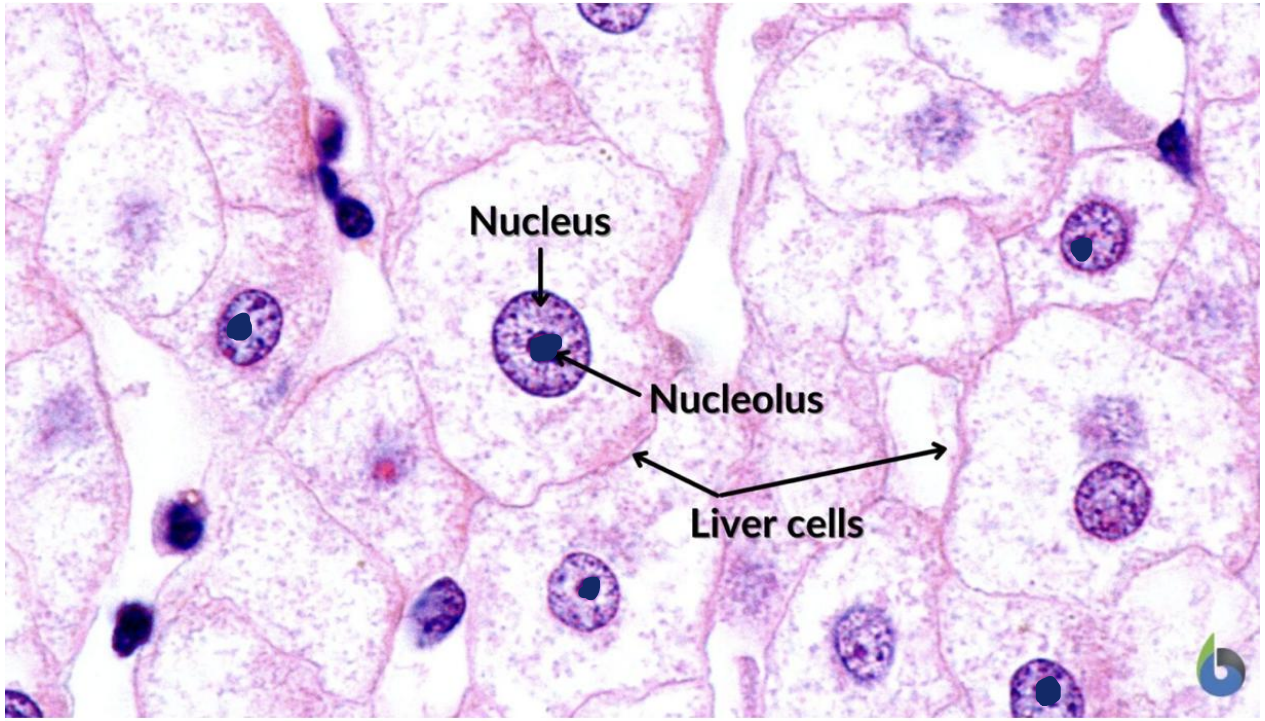


Euchromatin	Heterochromatin
✓ Extended finely granular material	✓ Coarse electron-dense material
✓ By L.M ➔ Pale basophilic	✓ By L.M ➔ Dark basophilic
✓ Carries active genes for protein synthesis	✓ Carries inactive genes
✓ Predominates in active cells	✓ Predominates in inactive cells

4-Nucleolus

■L.M:

- ➢Dark **basophilic**
- ➢May be **double** (according to cell's **activity**)



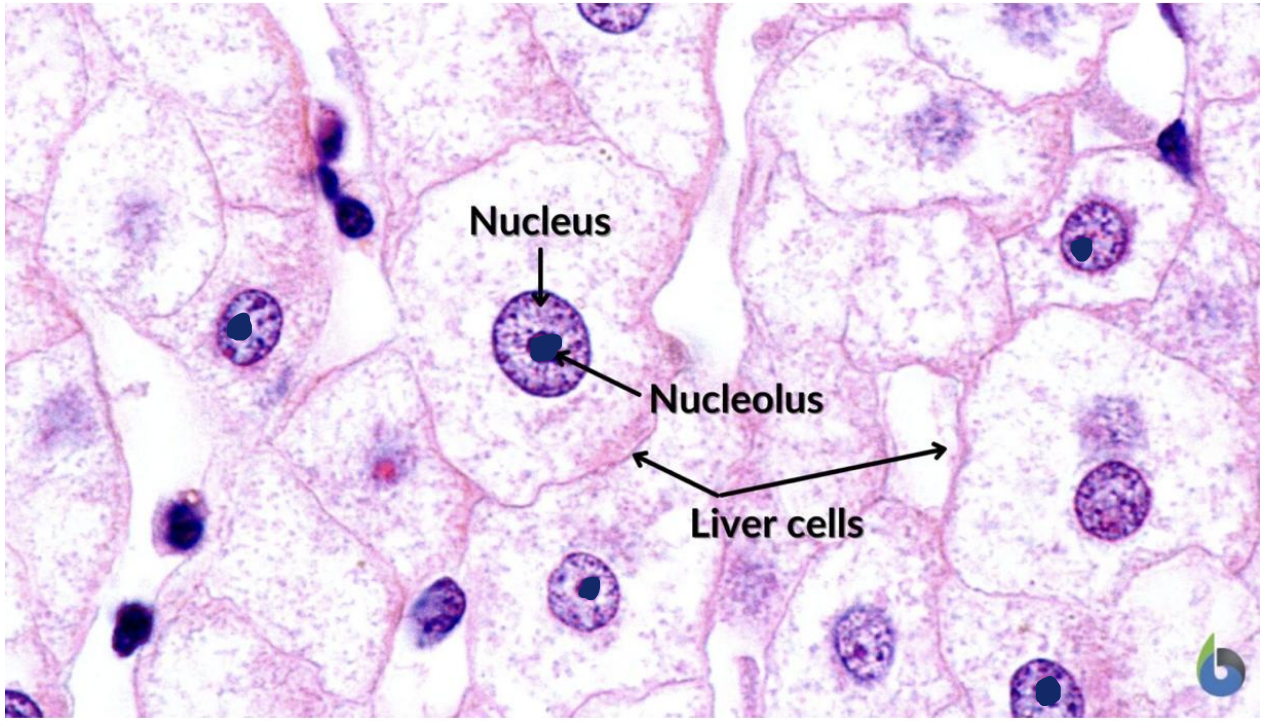
■E.M:

- ➢DNA filaments carrying rRNA genes

4-Nucleolus

■Function:

➢The site of formation of r-RNA, which means that the RNA parts of the ribosomes are produced here. They are combined later in the cytoplasm with proteins to form ribosomes (not in nucleolus!)



Functions of the Nucleus

1. Controls all cell functions

   [exclusively protein synthesis]

2. Storage of genetic information

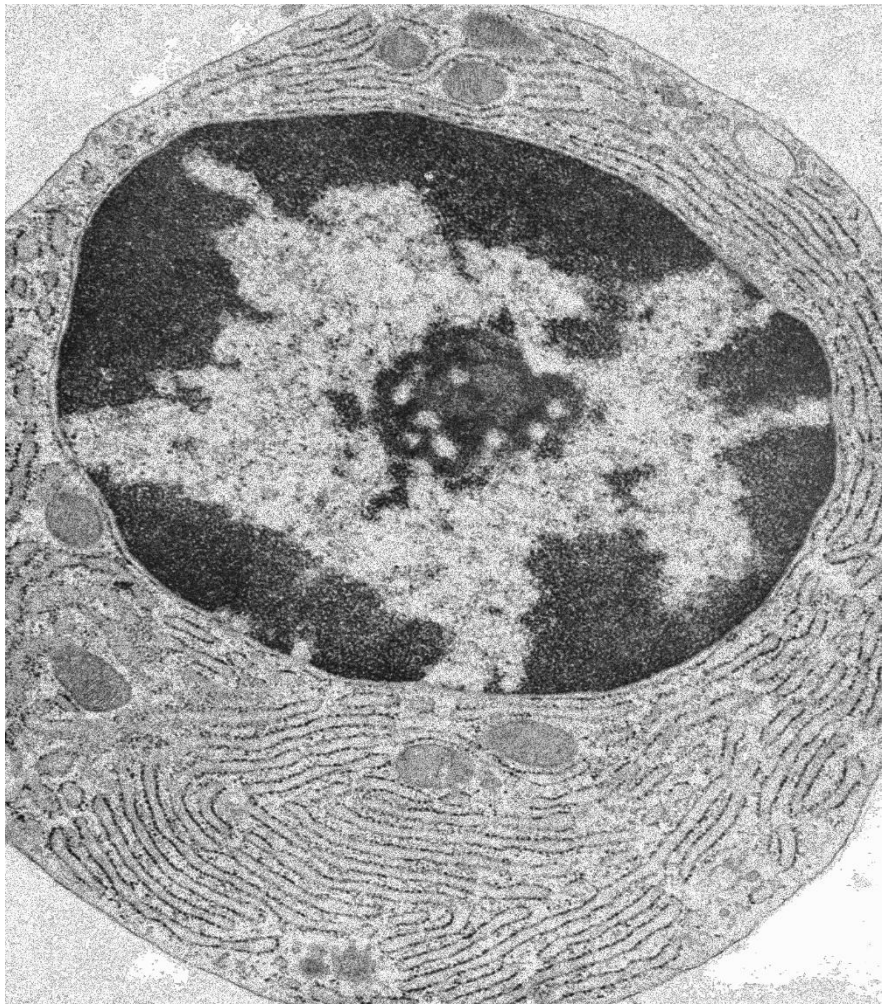
3. Formation of the three types of RNA

4. Controls cell division

5. Function of the nucleolus (look before)

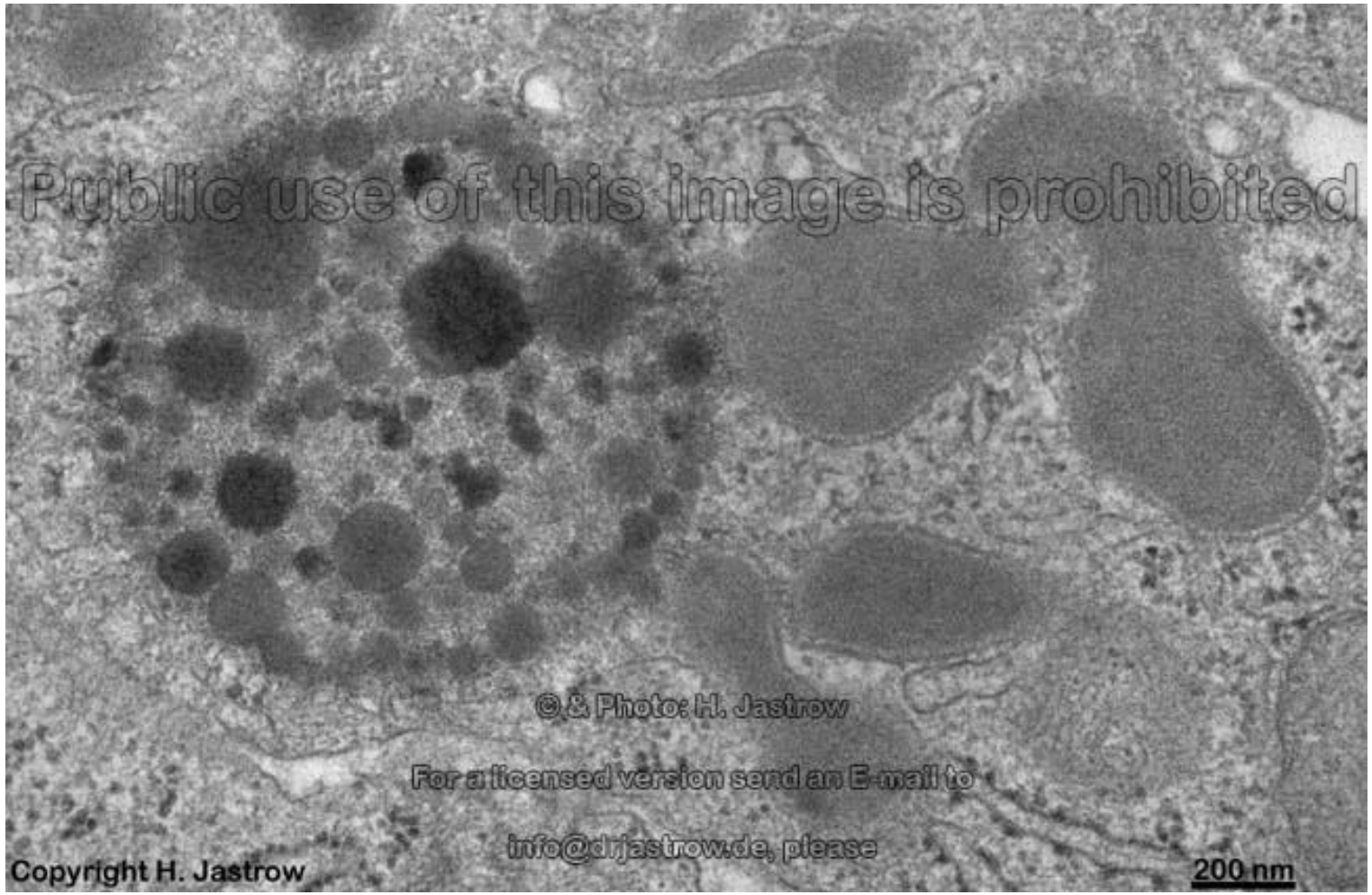


The Nucleus [E.M]



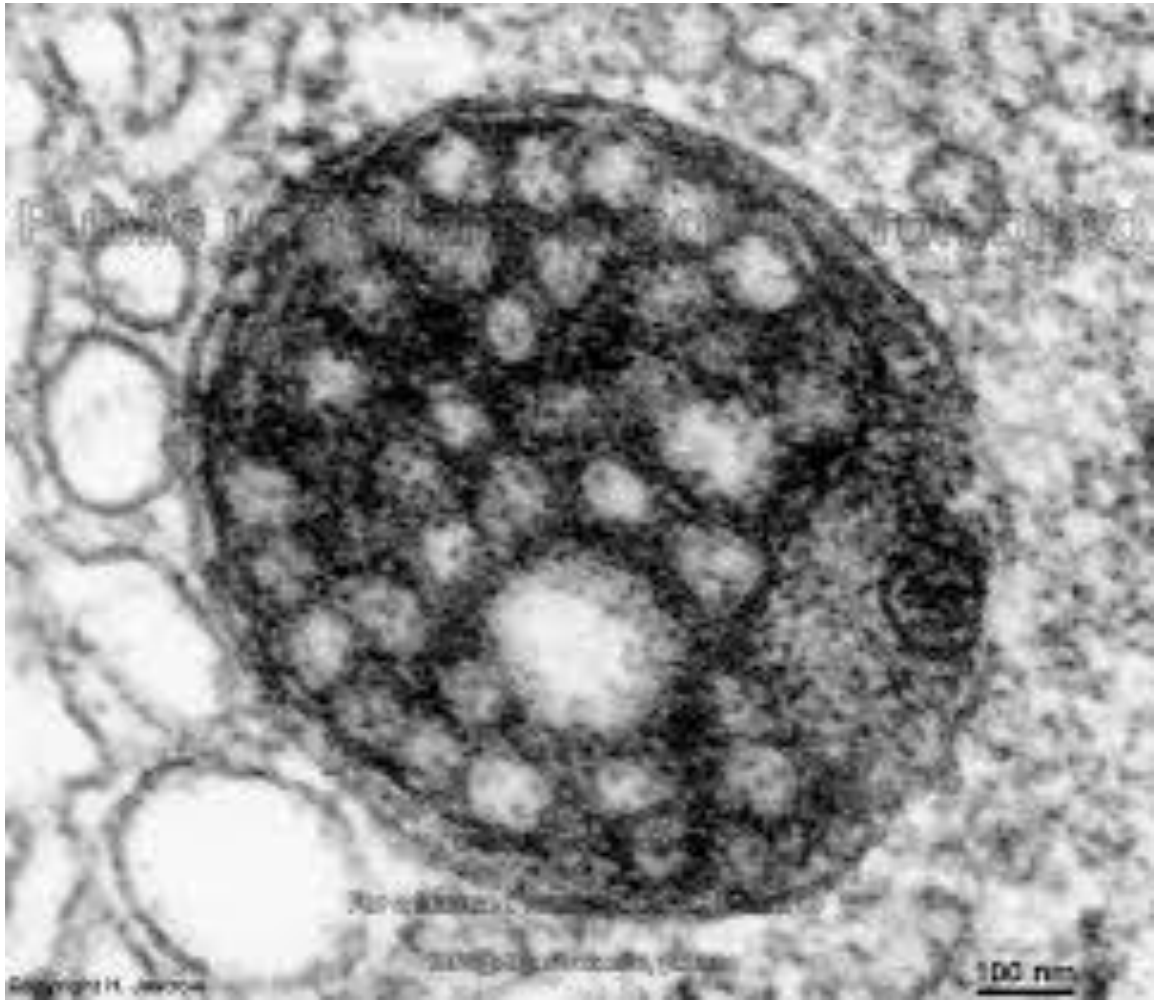
Practical slides of this session

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200 nm

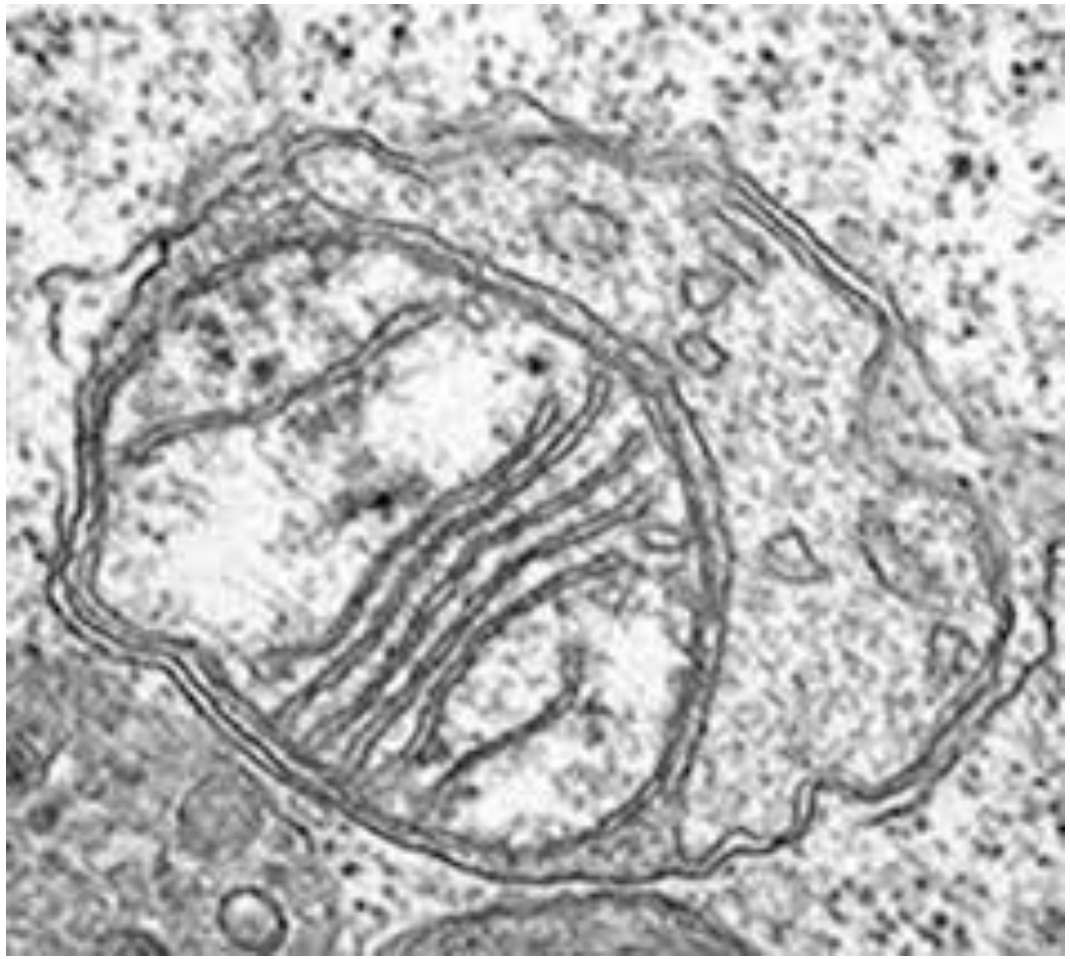
EM picture of a Heterolysosome

?


100 nm

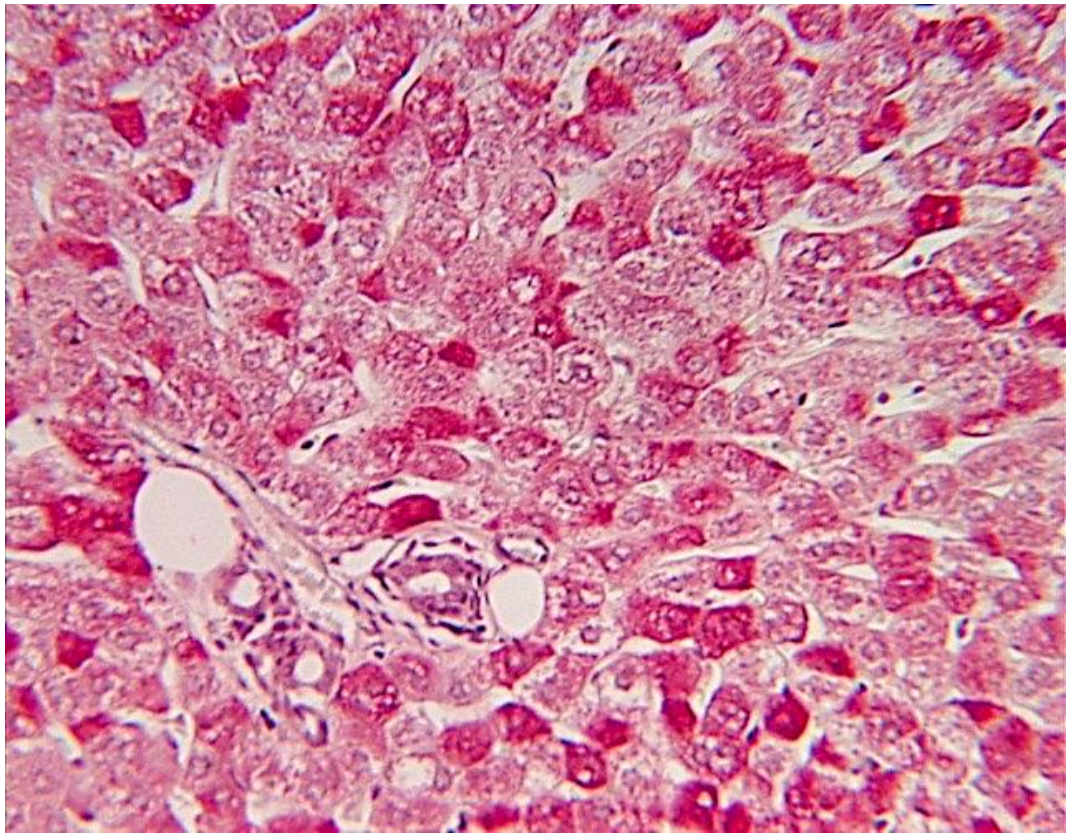
EM picture of a Multivesicular body

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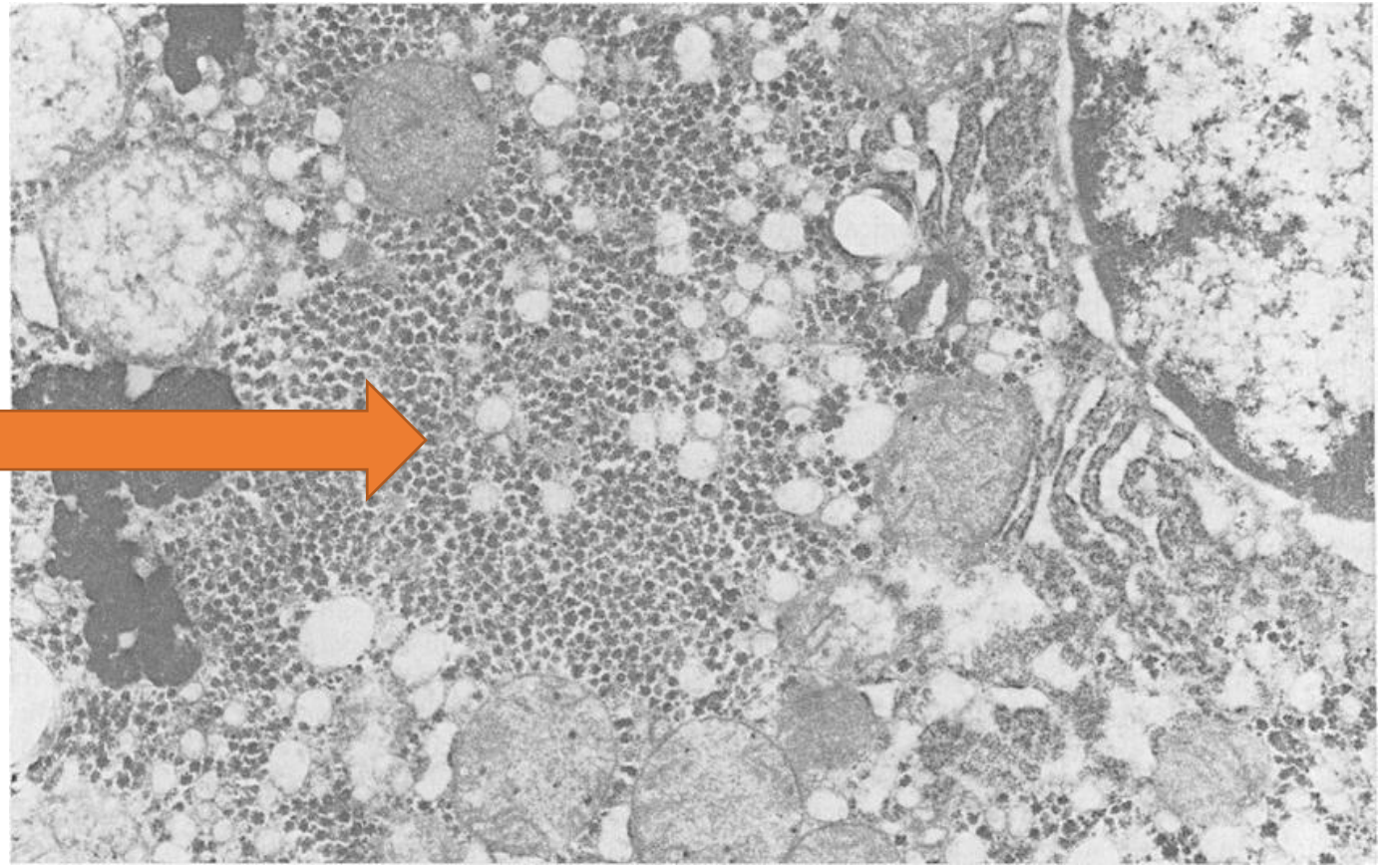
EM picture of an Autolysosome

?



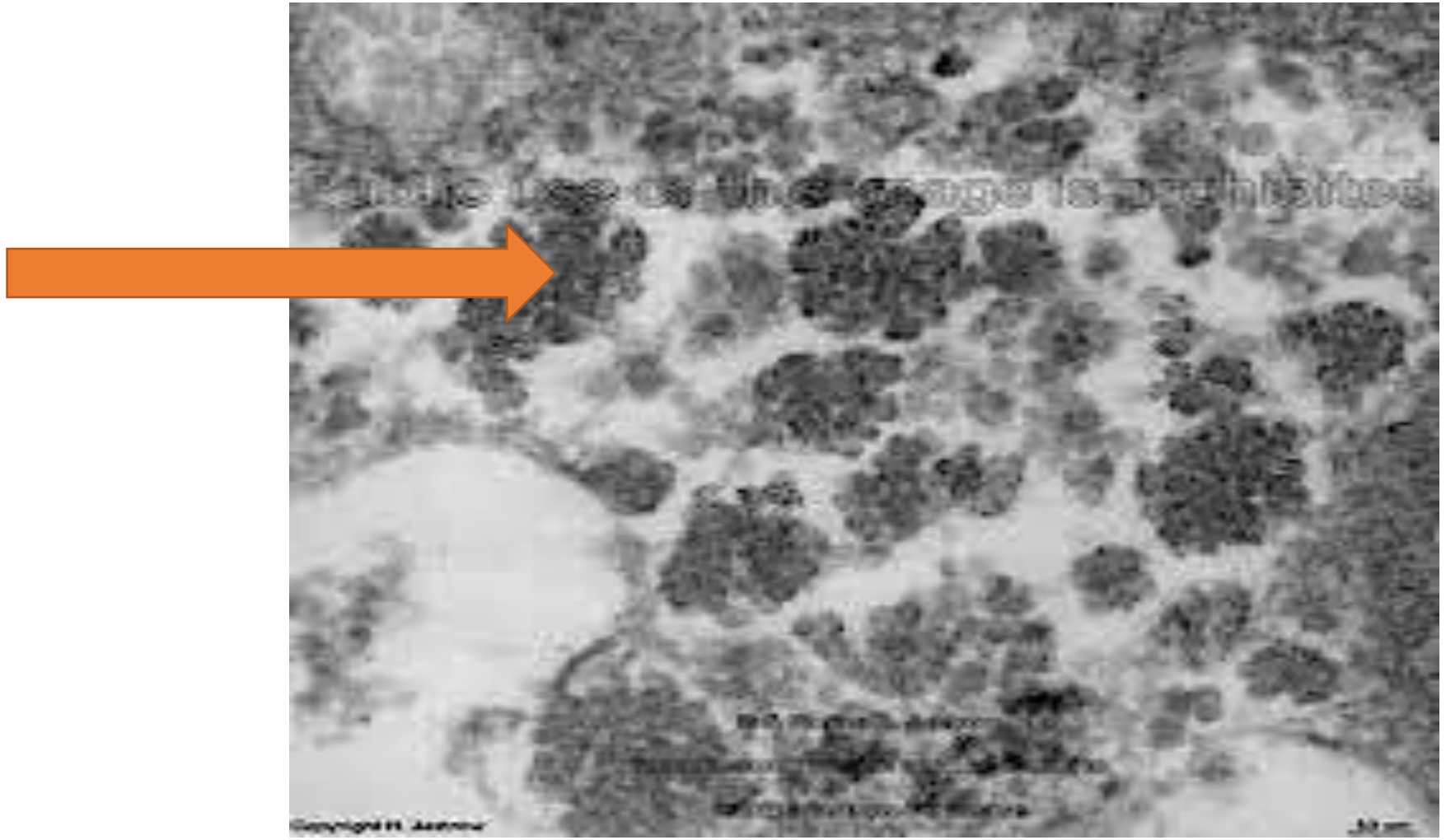
LM picture of glycogen stained with best carmine

?



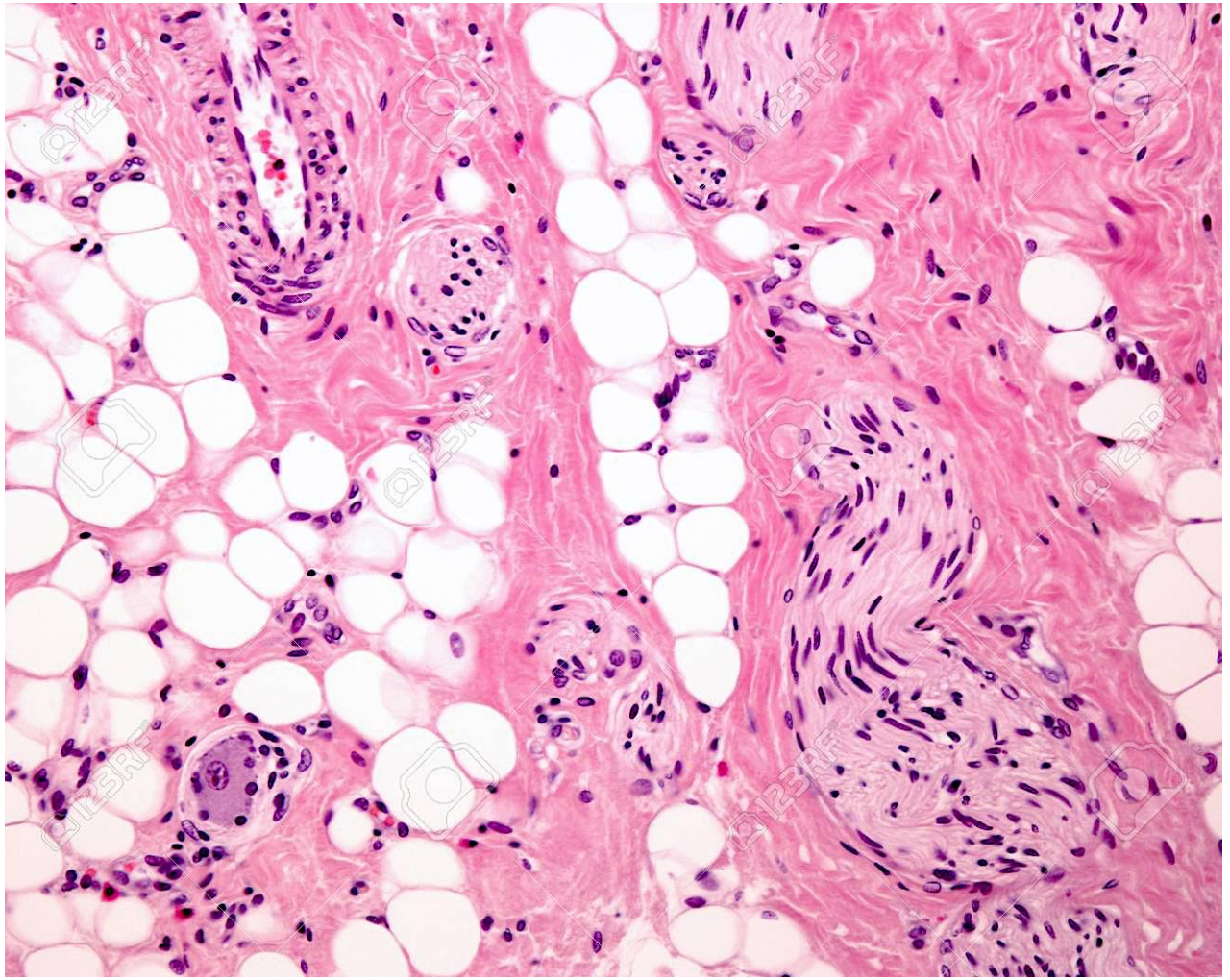
EM picture of α-glycogen

?



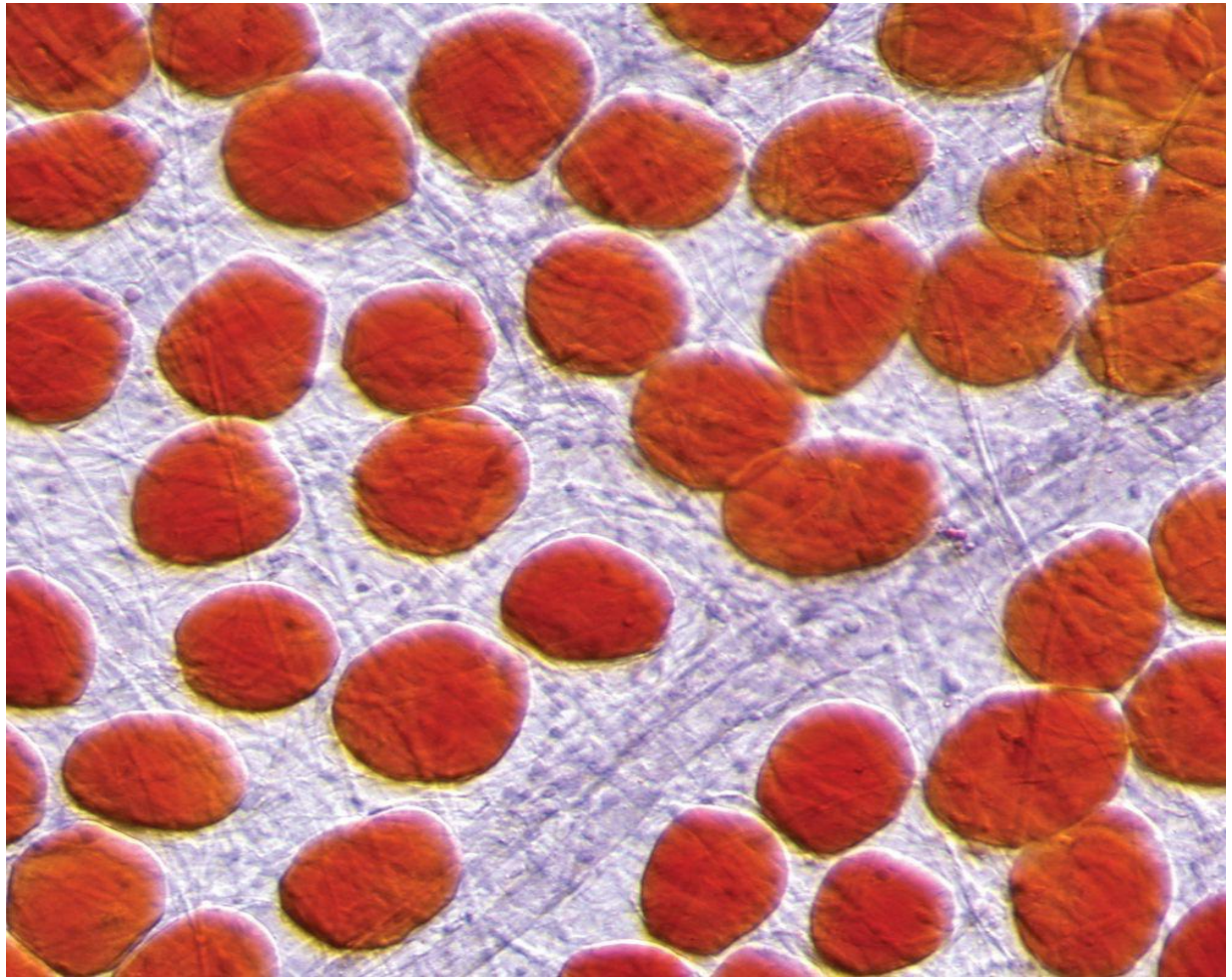
EM picture of β-glycogen (rosettes)

?



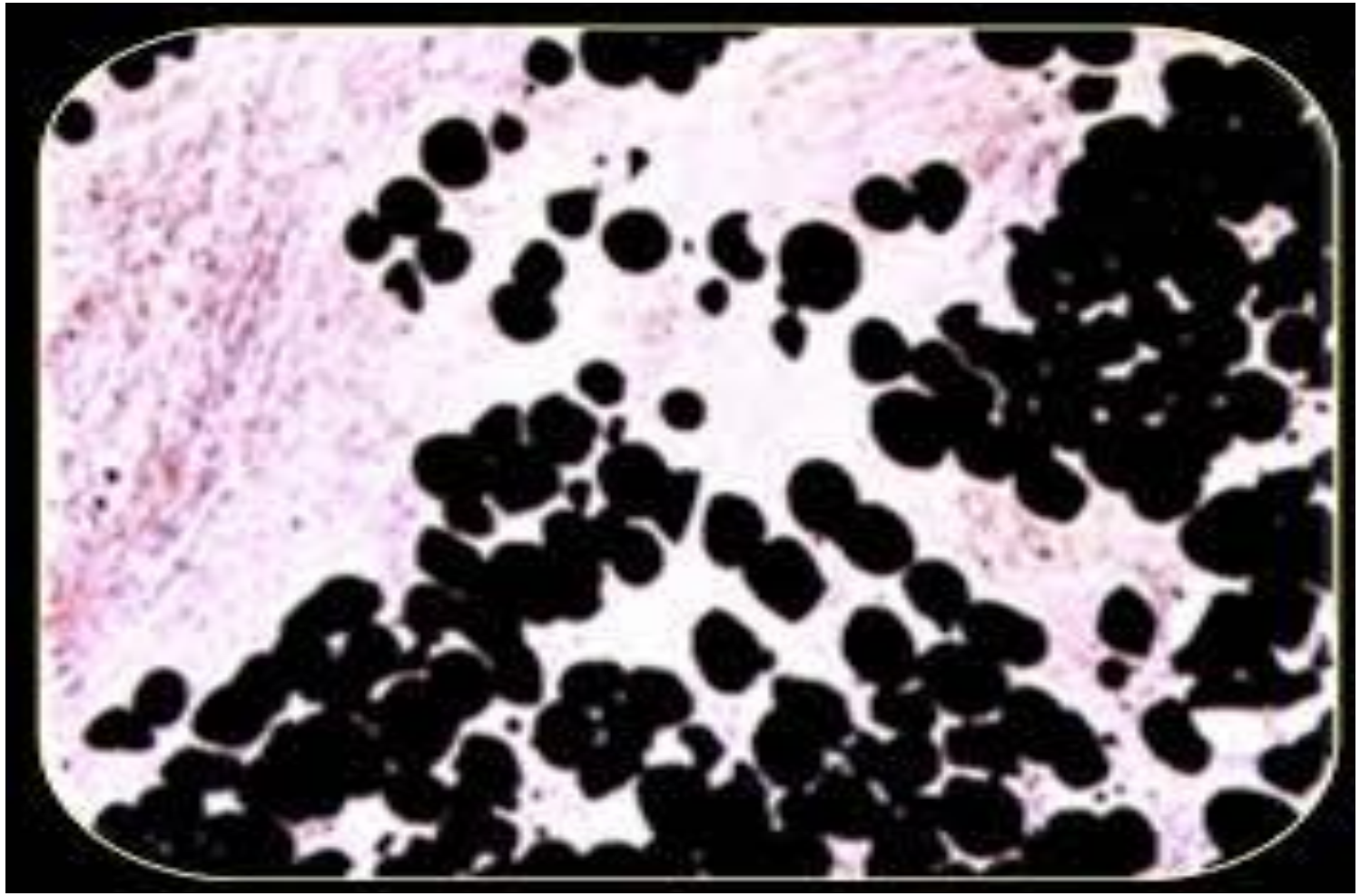
LM picture of fat cells

?



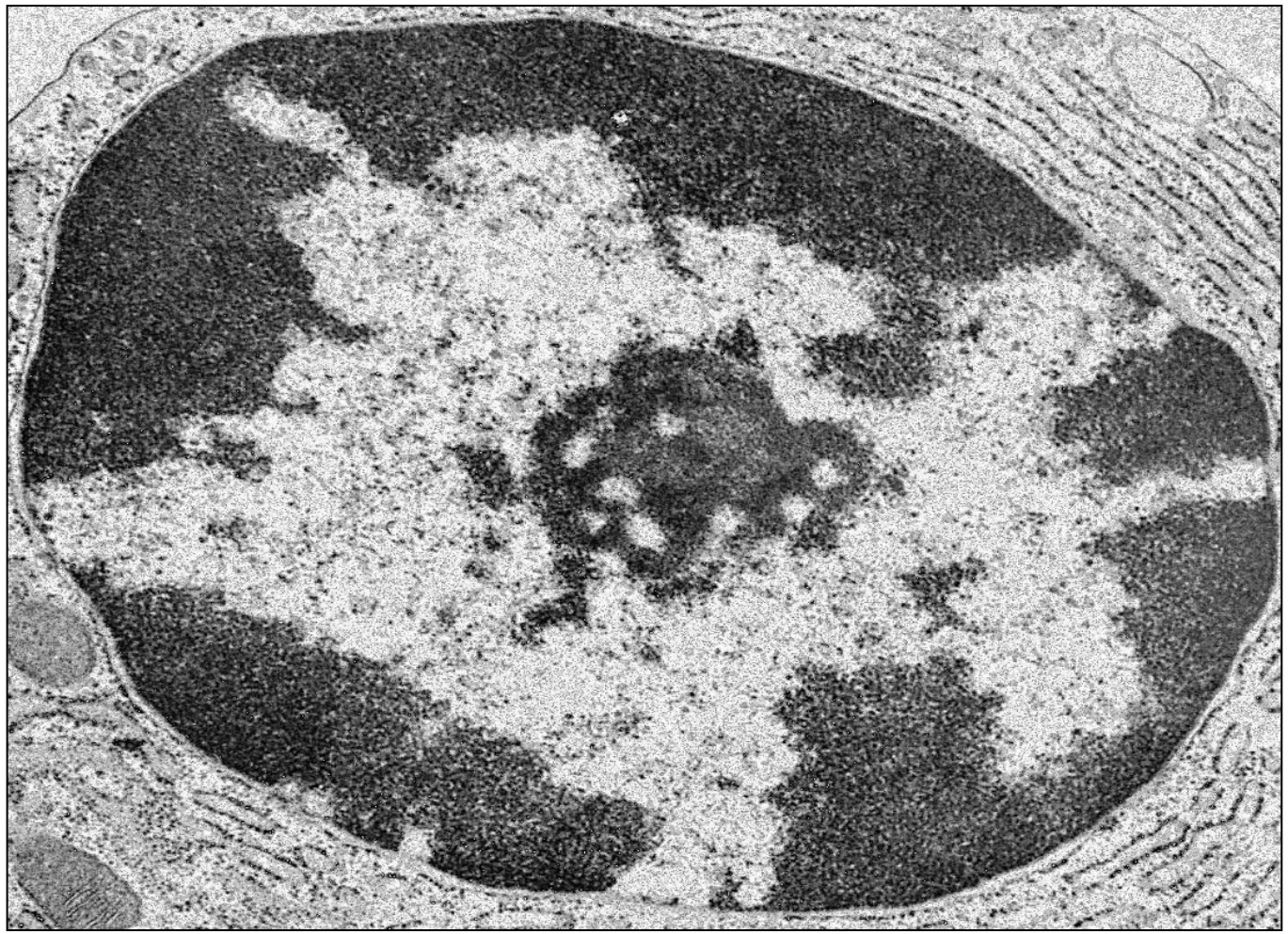
LM picture of fat cells stained with Sudan III

?



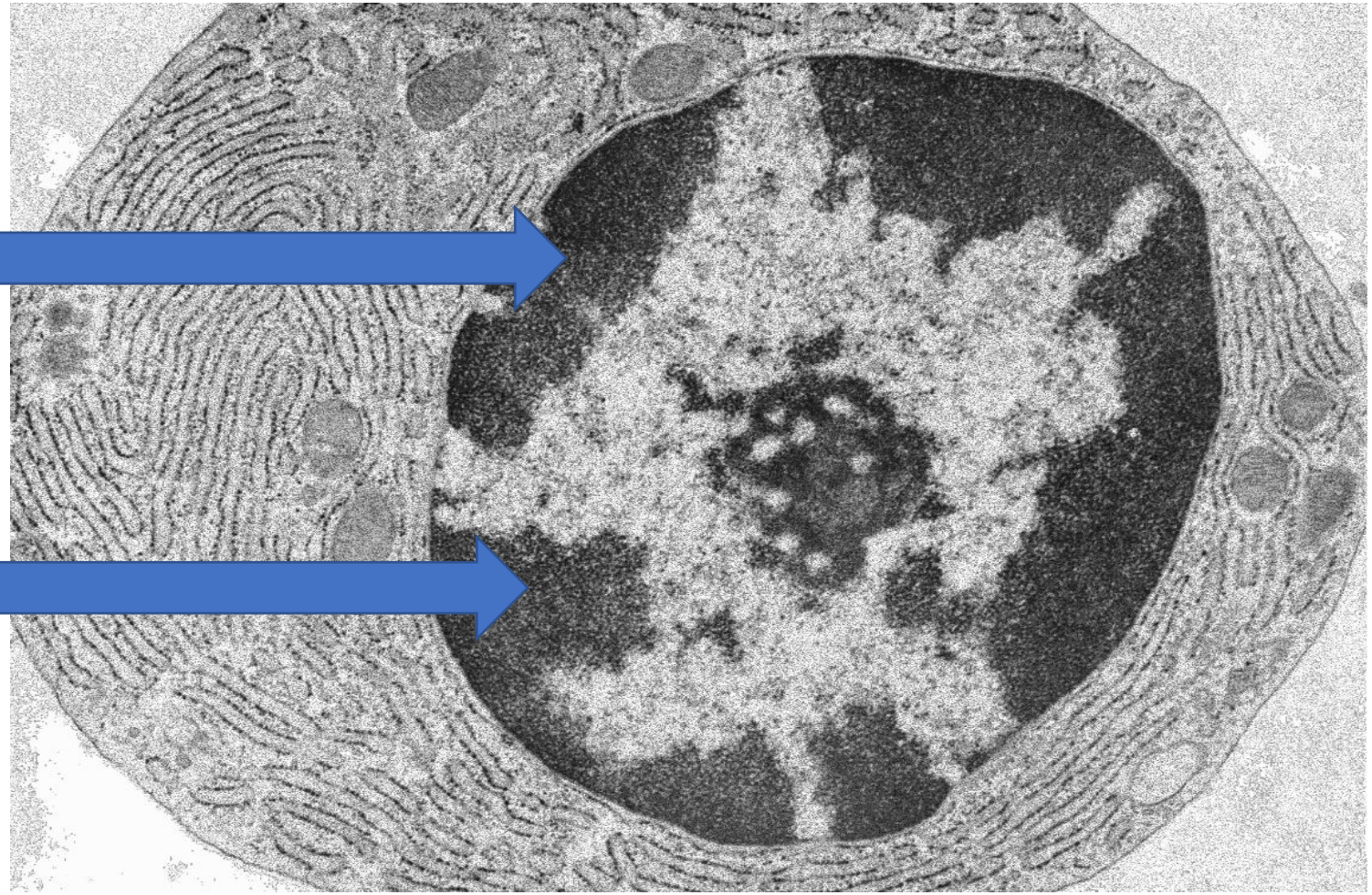
LM picture of fat cells stained with Sudan Black

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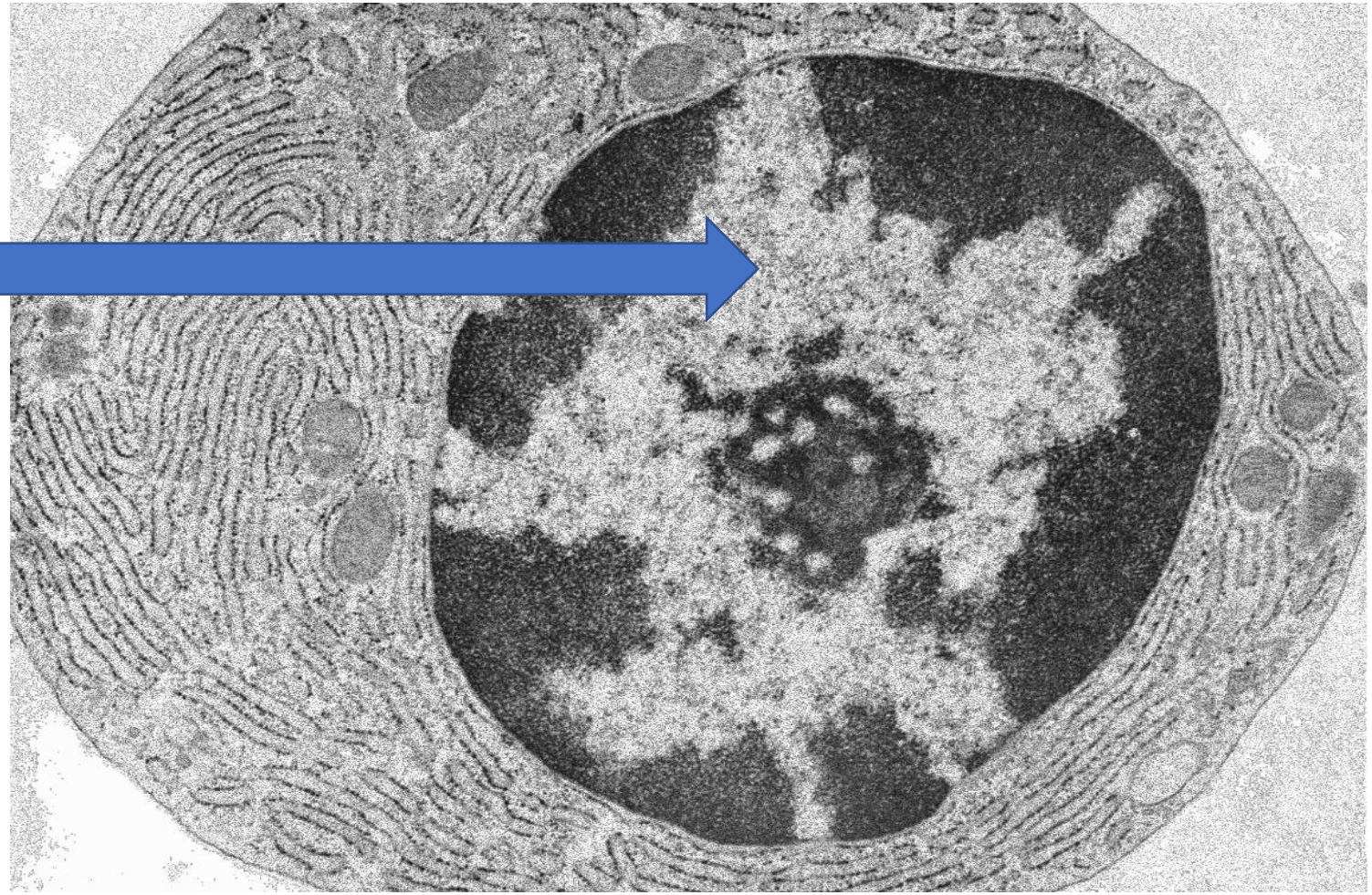
EM picture of a nucleus

?



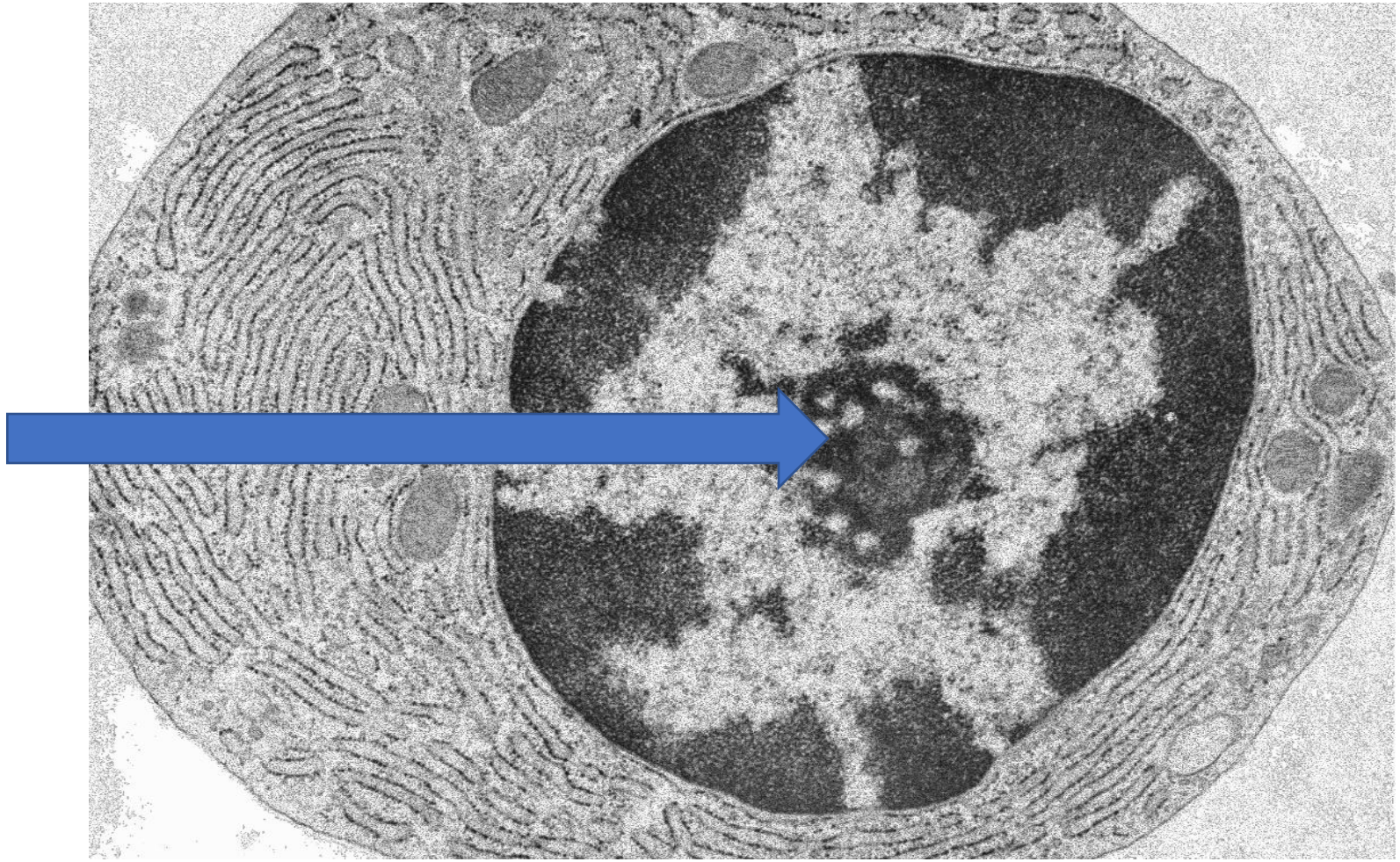
EM picture of Heterochromatin

?



EM picture of Euchromatin

?



EM picture of Nucleolus

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